

FURTHER TOPICS IN ANALYSIS

The broader perspective and additional tools made available through complex variable theory enable us to consider fruitfully a number of topics in analysis that have wide application in areas of relevance to physics. In this chapter we survey several such topics.

12.1 ORTHOGONAL POLYNOMIALS

Many physical problems lead to second-order differential equations corresponding to Sturm-Liouville problems, and often the solutions of interest in physics are polynomials, defined on a range and with weighting factors that make them eigenfunctions of Hermitian problems. A number of interesting features of such problems can be approached with the aid of complex variable theory.

Rodrigues Formulas

Odile Rodrigues showed that a large class of second-order Sturm-Liouville ordinary differential equations (ODEs) had polynomial solutions which could be put in a compact and useful form now generally called a Rodrigues formula. While such formulas could be presented case by case with an aura of coincidence or mystery, the approach we take here is to develop them from a general viewpoint, after which we can proceed to more detailed discussion of well-known special cases.

Consider a second-order Sturm-Liouville ODE of the general form

$$p(x)y'' + q(x)y' + \lambda y = 0, \quad (12.1)$$

with $p(x)$ and $q(x)$ restricted to the polynomial forms

$$p(x) = \alpha x^2 + \beta x + \gamma, \quad q(x) = \mu x + \nu. \quad (12.2)$$

The forms of p and q are sufficiently general to include most of the ODEs with classical sets of polynomials as solutions (the Legendre, Hermite, and Laguerre ODEs, among others). When Eq. (12.1) has as a solution a polynomial of degree n , we can write

$$y_n(x) = \sum_{j=0}^n g_j x^j, \quad (12.3)$$

with coefficient g_n nonzero. Setting to zero the coefficient of x^n when y_n is inserted into the ODE, we have

$$n(n-1)\alpha g_n + n\mu g_n + \lambda g_n = 0, \quad (12.4)$$

showing that the eigenvalue λ_n which corresponds to y_n must have the value

$$\lambda_n = -n(n-1)\alpha - n\mu. \quad (12.5)$$

In Chapter 7 we identified an ODE of the form of Eq. (12.1) as self-adjoint if $p'(x) = q(x)$, and also showed that if an ODE was not already self-adjoint as written, it could be converted to self-adjoint form by multiplying all its terms by a weight factor $w(x)$, which must be such that

$$(wp)' = wq, \quad \text{or} \quad w' = w \frac{q - p'}{p}. \quad (12.6)$$

As shown previously, this equation is separable and has solution

$$w(x) = p^{-1} \exp \left(\int^x \frac{q(x)}{p(x)} dx \right). \quad (12.7)$$

The introduction of w enables the ODE to assume the form

$$\frac{d}{dx} [w(x)p(x)y'] + \lambda w(x)y = 0, \quad (12.8)$$

which was useful for discussing orthogonality properties of its solutions.

Our current interest in $w(x)$, however, is in the observation by Rodrigues that its particular form permits the solutions $y_n(x)$ to be written in the compact and interesting form that is now called its **Rodrigues formula**:

$$y_n(x) = \frac{1}{w(x)} \left(\frac{d}{dx} \right)^n [wp(x)^n]. \quad (12.9)$$

The proof of Eq. (12.9) is both simple and ingenious. Using the defining condition for $w(x)$, Eq. (12.6), we first obtain

$$p[wp^n]' = wp^n[(n-1)p' + q]. \quad (12.10)$$

We then differentiate this equation $n + 1$ times and divide by w . Because p is only quadratic in x and q is linear, application of Leibniz's formula to the multiple differentiations leads to only three terms on the left-hand side and two on the right:

$$\begin{aligned} & \frac{p}{w} \left(\frac{d}{dx} \right)^{n+2} [wp^n] + \frac{(n+1)p'}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] + \frac{n(n+1)p''}{2w} \left(\frac{d}{dx} \right)^n [wp^n] \\ &= \frac{(n-1)p' + q}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] + \frac{(n+1)[(n-1)p'' + q']}{w} \left(\frac{d}{dx} \right)^n [wp^n]. \end{aligned} \quad (12.11)$$

Our objective is to manipulate Eq. (12.11) into a form showing that y_n as given in Eq. (12.9) is a solution to the ODE of Eq. (12.1). We start by identifying the terms with y_n where that is possible, and then, combining or canceling similar terms, we reach

$$\frac{p}{w} \left(\frac{d}{dx} \right)^{n+2} [wp^n] + \frac{2p' - q}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] - \left[\frac{n^2 - n - 2}{2} p'' + (n+1)q' \right] y_n = 0. \quad (12.12)$$

To complete our analysis we now need to move the factors $1/w$ so that only n differentiations appear to their right, enabling identification of the remaining terms of the equation with y_n or its derivatives. We note the identity

$$\begin{aligned} \frac{p}{w} \left(\frac{d}{dx} \right)^{n+2} [wp^n] &= p \left[\frac{1}{w} \left(\frac{d}{dx} \right)^n [wp^n] \right]'' - 2p \left(\frac{dw^{-1}}{dx} \right) \left(\frac{d}{dx} \right)^{n+1} [wp^n] \\ &\quad - p \left(\frac{d^2 w^{-1}}{dx^2} \right) \left(\frac{d}{dx} \right)^n [wp^n], \end{aligned}$$

which reduces, using Eq. (12.6), to

$$\frac{p}{w} \left(\frac{d}{dx} \right)^{n+2} [wp^n] = py_n'' + \frac{2(q - p')}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] - \left[p'' - q' - \frac{q - p'}{p} \right] y_n. \quad (12.13)$$

Substituting Eq. (12.13) into Eq. (12.12), some further simplification results:

$$py_n'' + \frac{q}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] - \left[\frac{n^2 - n}{2} p'' + nq' - \frac{q(q - p')}{p} \right] y_n = 0. \quad (12.14)$$

Our final step is to use the identity

$$\frac{q}{w} \left(\frac{d}{dx} \right)^{n+1} [wp^n] = qy_n' - \frac{q(q - p')}{p} y_n, \quad (12.15)$$

which brings us to

$$py_n'' + qy_n' - \left[\frac{n^2 - n}{2} p'' + nq' \right] y_n = 0. \quad (12.16)$$

Noting that $p'' = 2\alpha$ and $q' = \mu$, we confirm that y_n is a solution of Eq. (12.1) with the eigenvalue given in Eq. (12.5).

Finally, we need to show that Rodrigues' formula, Eq. (12.9), results in an expression that is a polynomial of degree n . We note that a typical term of that formula will involve a j -fold differentiation of w and an $(n - j)$ -fold differentiation of p^n . After the differentiation of p^n , we are left with p^j times a polynomial. The differentiation of w will, applying Eq. (12.6), leave (w/p^j) times a polynomial, and the numerator and denominator factors p^j cancel. In addition, the w from the differentiation cancels against the initial factor w^{-1} , leaving each term of y_n in polynomial form. When all terms of y_n are combined, the resulting polynomial must have the degree consistent with Eq. (12.5), namely n .

Example 12.1.1 RODRIGUES FORMULA FOR HERMITE ODE

The Hermite ODE is

$$y'' - 2xy' + \lambda y = 0, \quad \text{or} \quad py'' + qy' + \lambda y = 0$$

with $p = 1$, $q = -2x$. We easily find

$$w = \exp \left(\int^x (-2x) dx \right) = e^{-x^2}.$$

The Rodrigues formula is therefore (with a factor $(-1)^n$ to obtain the Hermite polynomials with their conventional signs)

$$y_n(x) = \frac{(-1)^n}{w} \left(\frac{d}{dx} \right)^n [wp^n] = (-1)^n e^{x^2} \left(\frac{d}{dx} \right)^n e^{-x^2}. \quad (12.17)$$

■

Schlaefli Integral

One of the nice features of the Rodrigues formulas is that the multiple differentiations can be converted to a convenient form by use of Cauchy's integral formula. Using Eq. (11.33), we have

$$y_n(x) = \frac{1}{w(x)} \frac{n!}{2\pi i} \oint_C \frac{w(z)[p(z)]^n}{(z-x)^{n+1}} dz, \quad (12.18)$$

where the contour C encloses the point x , and must be such that $w(z)[p(z)]^n$ is analytic everywhere on and within C . This formula is known as the **Schlaefli integral** for $y_n(x)$.

It is possible to introduce the Schlaefli integral as the definition of a set of functions y_n and, from that definition, prove that y_n is a solution to the corresponding ODE. Because we created the Schlaefli integral to represent a function already known to be a solution, verification that it solves the ODE becomes redundant.

Generating Functions

Many sets of functions arising in mathematical physics can be defined in terms of generating functions. Such functions include, but are not limited to the orthogonal polynomials y_n that have been the subject of our discussion of Rodrigues formulas. For now, we make no assumptions as to the source of the functions involved.

If $f_n(x)$ is a set of functions, defined for integer values of the index n , it may be the case that the $f_n(x)$ can be described as the coefficients of the powers of an auxiliary variable, t , in the expansion of a function $g(x, t)$, which is called a **generating function**:

$$g(x, t) = \sum_n c_n f_n(x) t^n. \quad (12.19)$$

The range of n may be semi-infinite, with $n \geq 0$, thereby describing a Taylor series, or it may extend from $-\infty$ to $+\infty$, thus describing a Laurent series. The additional coefficient, c_n , permits adjustment of the function set to an agreed-upon scaling. Different choices of c_n will also lead to different generating functions $g(x, t)$ for the same set of f_n .

Applying the residue theorem, we can see that the generating function expansion is closely related to contour integral representations of the functions f_n :

$$c_n f_n(x) = \frac{1}{2\pi i} \oint \frac{g(x, t)}{t^{n+1}} dt, \quad (12.20)$$

where the contour encircles $t = 0$ but no other singularities of the integrand (with respect to t).

A generating function may be regarded as providing the definition of a function set $f_n(x)$, or alternatively it may have been obtained as the encapsulation of the f_n which were already defined in some other way (e.g., as polynomial solutions of a Sturm-Liouville ODE). We shall later take up the issue of obtaining generating functions for previously specified f_n , focusing for now only on ways in which they can be used.

It is obvious that by explicitly evaluating the implied expansion one can extract the members of a function set from its generating function. However, a more important feature of generating functions is that they can be very useful in deriving relationships between members of the set f_n . For example,

$$\frac{\partial g(x, t)}{\partial t} = \sum_n n c_n f_n(x) t^{n-1} = \sum_n (n+1) c_{n+1} f_{n+1}(x) t^n,$$

and if we can relate g and $\partial g / \partial t$ we have a corresponding relation between f_n and f_{n+1} . Relations between the $f_n(x)$ and their derivatives $f'_n(x)$ can be deduced by differentiating $g(x, t)$ with respect to x .

Example 12.1.2 HERMITE POLYNOMIALS

A generating function formula for the Hermite polynomials $H_n(x)$ (at their conventional scaling) is

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}. \quad (12.21)$$

To develop a recurrence formula connecting H_n of contiguous index values, we compute

$$\frac{\partial}{\partial t} e^{-t^2+2tx} = (2x - 2t)e^{-t^2+2tx} = \sum_{n=0}^{\infty} n H_n(x) \frac{t^{n-1}}{n!}. \quad (12.22)$$

Expanding the exponential in the central member of Eq. (12.22) (and suppressing temporarily the argument of H_n),

$$\sum_{n=0}^{\infty} 2x H_n \frac{t^n}{n!} - \sum_{n=0}^{\infty} 2 H_n \frac{t^{n+1}}{n!} = \sum_{n=0}^{\infty} n H_n \frac{t^{n-1}}{n!}.$$

Extracting the coefficient of t^n from each of these summations, we reach (for each n)

$$\frac{2x H_n}{n!} - \frac{2 H_{n-1}}{(n-1)!} = \frac{(n+1) H_{n+1}}{(n+1)!},$$

which reduces to

$$2x H_n(x) - 2n H_{n-1}(x) = H_{n+1}(x). \quad (12.23)$$

Equation (12.23) is called a **recurrence formula**; it permits the construction of the entire series of H_n from starting values (typically H_0 and H_1 , which are easily computed directly).

A derivative formula can be obtained by differentiating Eq. (12.21) with respect to x . We have

$$\frac{\partial}{\partial x} e^{-t^2+2tx} = 2t e^{-t^2+2tx} = \sum_{n=0}^{\infty} H'_n(x) \frac{t^n}{n!}.$$

Substituting Eq. (12.21) into the central member of this equation, we get

$$\sum_{n=0}^{\infty} 2 H_n(x) \frac{t^{n+1}}{n!} = \sum_{n=0}^{\infty} H'_n(x) \frac{t^n}{n!},$$

which leads directly to

$$2n H_{n-1}(x) = H'_n(x). \quad (12.24)$$

■

In later chapters we illustrate the application of these ideas to a variety of special functions; in the next section of this chapter we apply them to a generating function that leads to quantities known as **Bernoulli numbers**.

Finding Generating Functions

To take generating functions out of the realm of magic, we next consider how they might be obtained. For a more or less arbitrary function set, this question has been a topic of current interest in mathematical research, with methods of several sorts devised during the past century by Rainville, Weisner, Truesdell, and others. See the works by McBride and Talman in Additional Readings.

For sets of polynomials arising in Sturm-Liouville problems and described by Rodrigues formulas, we can be more explicit. Using the Schlaefli integral, Eq. (12.18), we can form

$$g(x, t) = \frac{1}{w(x)} \sum_{n=0}^{\infty} c_n t^n \frac{n!}{2\pi i} \oint_C \frac{w(z)[p(z)]^n}{(z-x)^{n+1}} dz. \quad (12.25)$$

Recall that C encloses x and that wp^n must be analytic throughout the region within the contour.

In principle Eq. (12.25) can be evaluated to obtain $g(x, t)$, for example by choosing C to be such that the summation can be brought inside the z integral and (after specifying c_n) evaluating first the sum and then the contour integral. In practice the difficulty of doing this may depend on the problem, including the choice of c_n . We provide one example of the process.

Example 12.1.3 LEGENDRE POLYNOMIALS

We use the formal process described above to obtain a generating function for the Legendre polynomials. The Legendre ODE is of the form discussed in Eq. (12.1),

$$(1-x^2)y'' - 2xy' + \lambda y = 0,$$

implying that

$$p(x) = 1 - x^2, \quad q(x) = -2x,$$

and the equation is, as written, self-adjoint, so $w(x) = 1$. From the generating-function formula based on the Schlaefli integral, Eq. (12.25), we choose $c_n = (-1)^n/2^n n!$, thereby reaching

$$g(x, t) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n t^n}{2^n n!} \right) \frac{n!}{2\pi i} \oint_C \frac{(1-z^2)^n}{(z-x)^{n+1}} dz.$$

Interchanging the summation and integration (which we will justify later), the factors dependent on n form a geometric series, which we can sum:

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{(z^2-1)t}{2(z-x)} \right)^n \frac{1}{z-x} &= \frac{1}{z-x-\frac{1}{2}(z^2-1)t} \\ &= -\frac{2}{t} \left[z^2 - \frac{2z}{t} + \frac{2x-t}{t} \right]^{-1}. \end{aligned}$$

Inserting this result into the formula for $g(x, t)$, we now have

$$\begin{aligned} g(x, t) &= -\frac{2}{t} \frac{1}{2\pi i} \oint_C \left[z^2 - \frac{2z}{t} + \frac{2x-t}{t} \right]^{-1} dz \\ &= -\frac{2}{t} \frac{1}{2\pi i} \oint_C \frac{dz}{(z-z_1)(z-z_2)}, \end{aligned} \quad (12.26)$$

where z_1 and z_2 are the roots of the quadratic form in the first line of the equation:

$$z_1 = \frac{1}{t} - \frac{\sqrt{1 - 2xt + t^2}}{t}, \quad z_2 = \frac{1}{t} + \frac{\sqrt{1 - 2xt + t^2}}{t}.$$

In order for Eq. (12.26) to be valid, it must have been legitimate to interchange the summation and integration, which is the case only if the summation is uniformly convergent (with respect to z) for all points at which it is used (i.e., everywhere on the contour C). It is convenient to analyze the convergence for small t and x and for a contour with $|z| = 1$. Once a final formula has been obtained, its range of validity can be extended by appeal to analytic continuation.

On the assumed contour and for small x , there will be a range of $|t| \ll 1$ for which

$$\left| \frac{(z^2 - 1)t}{2(z - x)} \right| < 1,$$

guaranteeing convergence of the geometric series. We now return to the evaluation of the contour integral in Eq. (12.26). It has two poles, at $z = z_1$ and $z = z_2$. For small x and $|t|$, z_2 will be approximately $2/t$ and will be exterior to the contour, while z_1 will be close to the origin of z . Thus, only the residue of the integrand at $z = z_1$ will contribute to the contour integral, which will have the value

$$g(x, t) = -\frac{2}{t} \frac{1}{z_1 - z_2}.$$

Since

$$z_1 - z_2 = -\frac{2}{t} \sqrt{1 - 2xt + t^2},$$

we obtain the Legendre polynomial generating function as

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}}. \quad (12.27)$$

■

Summary—Orthogonal Polynomials

For five classical sets of orthogonal polynomials, we summarize in Table 12.1 their ODEs, Rodrigues formulas, and generating functions. Omitted from the list are important subsidiary polynomial sets (e.g., those connected with the associated Legendre and associated Laguerre ODEs).

Exercises

- 12.1.1** Starting from the Rodrigues formula in Table 12.1 for the Hermite polynomials H_n , derive the generating function for the H_n given in that table.

Table 12.1 Orthogonal Polynomials: ODEs, Rodrigues Formulas, and Generating Functions

Rodrigues Formula	Generating Function
Legendre: $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ $P_n(x) = \frac{1}{2^n n!} \left(\frac{d}{dx} \right)^n (x^2 - 1)^n$	$(1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) t^n$
Hermite: $y'' - 2xy' + 2ny = 0$ $H_n(x) = (-1)^n e^{x^2} \left(\frac{d}{dx} \right)^n e^{-x^2}$	$e^{-t^2 + 2xt} = \sum_{n=0}^{\infty} \frac{1}{n!} H_n(x) t^n$
Laguerre: $xy'' + (1-x)y' + ny = 0$ $L_n(x) = \frac{e^x}{n!} \left(\frac{d}{dx} \right)^n (x^n e^{-x})$	$\frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} L_n(x) t^n$
Chebyshev I: $(1-x^2)y'' - xy' + n^2y = 0$ $T_n(x) = \frac{(-1)^n (1-x^2)^{1/2}}{(2n-1)!!} \left(\frac{d}{dx} \right)^n (1-x^2)^{n-1/2}$	$\frac{1-t^2}{1-2xt+t^2} = T_0(x) + 2 \sum_{n=1}^{\infty} T_n(x) t^n$
Chebyshev II: $(1-x^2)y'' - 3xy' + n(n+2)y = 0$ $U_n(x) = \frac{(-1)^n (n+1)}{(2n+1)!! (1-x^2)^{1/2}} \left(\frac{d}{dx} \right)^n (1-x^2)^{n+1/2}$	$\frac{1}{1-2xt+t^2} = \sum_{n=0}^{\infty} U_n(x) t^n$

12.1.2 (a) Starting from the Laguerre ODE,

$$xy'' + (1-x)y' + \lambda y = 0,$$

obtain the Rodrigues formula for its polynomial solutions $L_n(x)$.

- (b) From the Rodrigues formula, scaled as in [Table 12.1](#), derive the generating function for the $L_n(x)$ given in that table.

12.1.3 Carry out in detail the steps needed to confirm that the $(n+1)$ -fold differentiation of [Eq. \(12.10\)](#) leads to [Eq. \(12.12\)](#).**12.1.4** Confirm the algebraic steps that convert [Eq. \(12.12\)](#) into [Eq. \(12.16\)](#).**12.1.5** Given the following integral representations, in which the contours encircle the origin but no other singular points, derive the corresponding generating functions:

- (a) Bessel functions:

$$J_n(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t-1/t)} t^{-n-1} dt.$$

- (b) Modified Bessel functions:

$$I_n(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t+1/t)} t^{-n-1} dt.$$

- 12.1.6** Expand the generating function for the Legendre polynomials, $(1 - 2tz + t^2)^{-1/2}$, in powers of t . Assume that t is small. Collect the coefficients of t^0 , t^1 , and t^2 .

$$\begin{aligned} \text{ANS. } a_0 &= P_0(z) = 1, \\ a_1 &= P_1(z) = z, \\ a_2 &= P_2(z) = \frac{1}{2}(3z^2 - 1). \end{aligned}$$

- 12.1.7** The set of Chebyshev polynomials usually denoted $U_n(x)$ has the generating-function formula

$$\frac{1}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} U_n(x)t^n.$$

Derive a recurrence formula (for integer $n \geq 0$) connecting three U_n of consecutive n .

12.2 BERNOULLI NUMBERS

A generating-function approach is a convenient way to introduce the set of numbers first used in mathematics by Jacques (James, Jacob) Bernoulli. These quantities have been defined in a number of different ways, so extreme care must be taken in combining formulas from works by different authors. Our definition corresponds to that used in the reference work *Handbook of Mathematical Functions* (AMS-55). See Additional Readings.

Since the Bernoulli numbers, denoted B_n , do not depend on a variable, their generating function depends only on a single (complex) variable, and the generating-function formula has the specific form

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} \frac{B_n t^n}{n!}. \quad (12.28)$$

The inclusion of the factor $1/n!$ in the definition is just one of the ways some definitions of Bernoulli numbers differ. We defer for the moment the important question as to the circle of convergence of the expansion in Eq. (12.28).

Since Eq. (12.28) is a Taylor series, we may identify the B_n as successive derivatives of the generating function:

$$B_n = \left[\frac{d^n}{dt^n} \left(\frac{t}{e^t - 1} \right) \right]_{t=0}. \quad (12.29)$$

To obtain B_0 , we must take the limit of $t/(e^t - 1)$ as $t \rightarrow 0$, easily finding $B_0 = 1$. Applying Eq. (12.29), we also have

$$B_1 = \left. \frac{d}{dt} \left(\frac{t}{e^t - 1} \right) \right|_{t=0} = \lim_{t \rightarrow 0} \left(\frac{1}{e^t - 1} - \frac{te^t}{(e^t - 1)^2} \right) = -\frac{1}{2}. \quad (12.30)$$

In principle we could continue to obtain further B_n , but it is more convenient to proceed in a more sophisticated fashion. Our starting point is to examine

$$\begin{aligned}\sum_{n=2}^{\infty} \frac{B_n t^n}{n!} &= \frac{t}{e^t - 1} - B_0 - B_1 t = \frac{t}{e^t - 1} - 1 + \frac{t}{2} \\ &= \frac{-t}{e^{-t} - 1} - 1 - \frac{t}{2},\end{aligned}\quad (12.31)$$

where we have used the fact that

$$\frac{t}{e^t - 1} = \frac{-t}{e^{-t} - 1} - t. \quad (12.32)$$

Equation (12.31) shows that the summation on its left-hand side is an even function of t , leading to the conclusion that all B_n of odd n (other than B_1) must vanish.

We next use the generating function to obtain a recursion relation for the Bernoulli numbers. We form

$$\begin{aligned}\frac{e^t - 1}{t} \frac{t}{e^t - 1} &= 1 = \left[\sum_{m=0}^{\infty} \frac{t^m}{(m+1)!} \right] \left[1 - \frac{t}{2} + \sum_{n=1}^{\infty} B_{2n} \frac{t^{2n}}{(2n)!} \right] \\ &= 1 + \sum_{m=1}^{\infty} t^m \left[\frac{1}{(m+1)!} - \frac{1}{2m!} \right] \\ &\quad + \sum_{N=2}^{\infty} t^N \sum_{n=1}^{\leq N/2} \frac{B_{2n}}{(2n)!(N-2n+1)!} \\ &= 1 + \sum_{N=2}^{\infty} \frac{t^N}{(N+1)!} \left[-\frac{N-1}{2} + \sum_{n=1}^{\leq N/2} \binom{N+1}{2n} B_{2n} \right].\end{aligned}\quad (12.33)$$

Since the coefficient of each power of t in the final summation of Eq. (12.33) must vanish, we may set to zero for each N the expression in its square brackets. Changing N , if even, to $2N$ and if odd, to $2N-1$, Eq. (12.33) leads to the pair of equations

$$\begin{aligned}N - \frac{1}{2} &= \sum_{n=1}^N \binom{2N+1}{2n} B_{2n}, \\ N - 1 &= \sum_{n=1}^{N-1} \binom{2N}{2n} B_{2n}.\end{aligned}\quad (12.34)$$

Either of these equations can be used to obtain the B_{2n} sequentially, starting from B_2 . The first few B_n are listed in Table 12.2.

To obtain additional relations involving the Bernoulli numbers, we next consider the following representation of $\cot t$:

$$\cot t = \frac{\cos t}{\sin t} = i \left(\frac{e^{it} + e^{-it}}{e^{it} - e^{-it}} \right) = i \left(\frac{e^{2it} + 1}{e^{2it} - 1} \right) = i \left(1 + \frac{2}{e^{2it} - 1} \right).$$

Table 12.2 Bernoulli Numbers

n	B_n	B_n
0	1	1.000000000
1	$-\frac{1}{2}$	-0.500000000
2	$\frac{1}{6}$	0.166666667
4	$-\frac{1}{30}$	-0.033333333
6	$\frac{1}{42}$	0.023809524
8	$-\frac{1}{30}$	-0.033333333
10	$\frac{5}{66}$	0.075757576

Note. Further values are given in AMS-55;
see Abramowitz in Additional Readings.

Multiplying by t and rearranging slightly,

$$\begin{aligned}
 t \cot t &= \frac{2it}{2} + \frac{2it}{e^{2it} - 1} = \sum_{n=0}^{\infty} B_{2n} \frac{(2it)^{2n}}{(2n)!} \\
 &= \sum_{n=0}^{\infty} (-1)^n B_{2n} \frac{(2t)^{2n}}{(2n)!}, \tag{12.35}
 \end{aligned}$$

where the term $2it/2$ has canceled the B_1 term that would otherwise appear in the expansion.

Now that we have our Bernoulli-number expansion identified with $t \cot t$, we can see that it represents a function with singularities (poles) at $t = m\pi$, where $m = \pm 1, \pm 2, \dots$. There is no singularity at $t = 0$ (due to the presence of the factor t), so the singularity nearest the expansion point (the origin) is at $|t| = \pi$. Since the argument in the expansion is $2t$, we conclude that the generating series for the Bernoulli numbers, Eq. (12.28), will have the radius of convergence $|2t| = 2\pi$. This observation is, of course, consistent with the fact that the zeros of $e^t - 1$ are for t at integer multiples of $2\pi i$.

To obtain another representation of the Bernoulli numbers, we write B_n using the contour-integration formula, Eq. (12.20). Noting that for use in this equation $c_n f_n(x) = B_n/n!$, we have

$$B_n = \frac{n!}{2\pi i} \oint \frac{t}{e^t - 1} \frac{dt}{t^{n+1}}, \tag{12.36}$$

where the integral is a circle within the radius of convergence of the generating series. We can, at least in principle, evaluate the integral using the residue theorem. For $n = 0$ we have a simple pole with a residue of $+1$, and

$$B_0 = \frac{0!}{2\pi i} \cdot 2\pi i(+1) = 1.$$

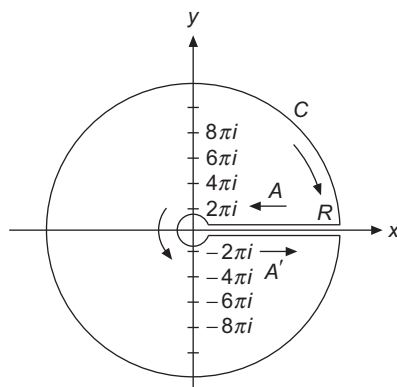


FIGURE 12.1 Contour of integration for Bernoulli numbers.

For $n = 1$ the singularity at $t = 0$ becomes a second-order pole, and the limiting process prescribed by Eq. (11.68) yields the residue $-\frac{1}{2}$, so

$$B_1 = \frac{1!}{2\pi i} \cdot 2\pi i \left(-\frac{1}{2} \right) = -\frac{1}{2},$$

consistent with our previous result. For $n \geq 2$ the poles at $t = 0$ are of increasing order and this procedure becomes rather tedious, so we resort to a different approach. We deform the contour of our integral representation as shown in Fig. 12.1, which differs from the original circular contour in that it surrounds all the poles of the integrand at $t = \pm 2\pi mi$, $m = 1, 2, \dots$, while avoiding the inclusion of the pole at $t = 0$. In contrast to the high-order pole at $t = 0$, the other poles are all first-order, with residues that are easily evaluated.

To use the new contour, we need to identify the contributions from its constituent parts. The direction of travel around the contour causes the small circle about $t = 0$ to contribute $+2\pi i$ times the residue of the integrand at $t = 0$, i.e., the result that when multiplied by $n!/2\pi i$ is equal to B_n . The remainder of the contour makes no contribution to the integral: (1) Because the integrand is analytic along the real axis and there is no branch cut there, the segments A and A' , which are in opposite directions of travel, cancel; and (2) the large circle contributes negligibly (for $n \geq 2$) because at large $|t|$ the integrand behaves asymptotically as $1/|t|^n$. Noting that the poles at nonzero t are encircled in a clockwise sense, we have the following relatively simple result (for $n \geq 2$):

$$B_n = -\frac{n!}{2\pi i} \sum 2\pi i \left(\text{residues of } \frac{t^{-n}}{e^t - 1} \text{ at poles } t \neq 0 \right). \quad (12.37)$$

Since the residue at $t = 2\pi mi$ is simply $(2\pi mi)^{-n}$, Eq. (12.37) becomes

$$B_n = -\frac{n!}{(2\pi i)^n} \sum_{m=1}^{\infty} \left[\frac{1}{m^n} + \frac{1}{(-m)^n} \right],$$

which further reduces, for $2n \geq 2$, to

$$B_{2n} = (-1)^{n+1} \frac{(2n)!}{(2\pi)^{2n}} \sum_{m=1}^{\infty} \frac{2}{m^{2n}} = (-1)^{n+1} \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n), \quad (12.38)$$

$$B_{2n+1} = 0.$$

Note that the B_n of odd $n > 1$ are correctly shown to vanish, and that the Bernoulli numbers of even $n > 0$ are identified as proportional to Riemann zeta functions, which first appeared in this book at Eq. (1.12). We repeat the definition:

$$\zeta(z) = \sum_{m=1}^{\infty} \frac{1}{m^z}.$$

Equation (12.38) is an important result because we already have a straightforward way to obtain values of the B_n , via Eq. (12.34), and Eq. (12.38) can be inverted to give a closed expression for $\zeta(2n)$, which otherwise was known only as a summation. This representation of the Bernoulli numbers was discovered by Euler.

It is readily seen from Eq. (12.38) that $|B_{2n}|$ increases without limit as $n \rightarrow \infty$. Numerical values have been calculated by Glaisher.¹ Illustrating the divergent behavior of the Bernoulli numbers, we have

$$B_{20} = -5.291 \times 10^2$$

$$B_{200} = -3.647 \times 10^{215}.$$

Some authors prefer to define the Bernoulli numbers with a modified version of Eq. (12.38) by using

$$\mathcal{B}_n = \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n), \quad (12.39)$$

the subscript being just half of our subscript and all signs positive. Again, when using other texts or references, you must check to see exactly how the Bernoulli numbers are defined.

The Bernoulli numbers occur frequently in number theory. The von Staudt-Clausen theorem states that

$$B_{2n} = A_n - \frac{1}{p_1} - \frac{1}{p_2} - \frac{1}{p_3} - \cdots - \frac{1}{p_k}, \quad (12.40)$$

in which A_n is an integer and $p_1, p_2, \dots, p_k$ are all the prime numbers such that $p_i - 1$ is a divisor of $2n$. It may readily be verified that this holds for

$$B_6 \quad (A_3 = 1, \quad p = 2, 3, 7),$$

$$B_8 \quad (A_4 = 1, \quad p = 2, 3, 5),$$

$$B_{10} \quad (A_5 = 1, \quad p = 2, 3, 11),$$

and other special cases.

¹J. W. L. Glaisher, table of the first 250 Bernoulli numbers (to nine figures) and their logarithms (to ten figures). *Trans. Cambridge Philos. Soc.* **12**: 390 (1871-1879).

The Bernoulli numbers appear in the summation of integral powers of the integers,

$$\sum_{j=1}^N j^p, \quad p \text{ integral,}$$

and in numerous series expansions of the transcendental functions, including $\tan x$, $\cot x$, $\ln |\sin x|$, $(\sin x)^{-1}$, $\ln |\cos x|$, $\ln |\tan x|$, $(\cosh x)^{-1}$, $\tanh x$, and $\coth x$. For example,

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots + \frac{(-1)^{n-1}2^{2n}(2^{2n}-1)B_{2n}}{(2n)!}x^{2n-1} + \cdots. \quad (12.41)$$

The Bernoulli numbers are likely to appear in such series expansions because of the definition, Eq. (12.28), the form of Eq. (12.35), and the relation to the Riemann zeta function, Eq. (12.38).

Bernoulli Polynomials

If Eq. (12.28) is generalized slightly, we have

$$\frac{te^{ts}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(s) \frac{t^n}{n!} \quad (12.42)$$

defining the **Bernoulli polynomials**, $B_n(s)$. It is clear that $B_n(s)$ will be a polynomial of degree n , since the Taylor expansion of the generating function will contain contributions in which each instance of t may (or may not) be accompanied by a factor s . The first seven Bernoulli polynomials are given in Table 12.3.

If we set $s = 0$ in the generating function formula, Eq. (12.42), we have

$$B_n(0) = B_n, \quad n = 0, 1, 2, \dots, \quad (12.43)$$

showing that the Bernoulli polynomial evaluated at zero equals the corresponding Bernoulli number.

Table 12.3 Bernoulli Polynomials

$B_0 = 1$
$B_1 = x - \frac{1}{2}$
$B_2 = x^2 - x + \frac{1}{6}$
$B_3 = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$
$B_4 = x^4 - 2x^3 + x^2 - \frac{1}{30}$
$B_5 = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x$
$B_6 = x^6 - 3x^5 + \frac{5}{2}x^4 - \frac{1}{2}x^2 + \frac{1}{42}$

Two other important properties of the Bernoulli polynomials follow from the defining relation, Eq. (12.42). If we differentiate both sides of that equation with respect to s , we have

$$\begin{aligned}\frac{t^2 e^{ts}}{e^t - 1} &= \sum_{n=0}^{\infty} B'_n(s) \frac{t^n}{n!} \\ &= \sum_{n=0}^{\infty} B_n(s) \frac{t^{n+1}}{n!} = \sum_{n=1}^{\infty} B_{n-1}(s) \frac{t^n}{(n-1)!},\end{aligned}\quad (12.44)$$

where the second line of Eq. (12.44) is obtained by rewriting its left-hand side using the generating-function formula. Equating the coefficients of equal powers of t in the two lines of Eq. (12.44), we obtain the differentiation formula

$$\frac{d}{ds} B_n(s) = n B_{n-1}(s), \quad n = 1, 2, 3, \dots \quad (12.45)$$

We also have a symmetry relation, which we can obtain by setting $s = 1$ in Eq. (12.42). The left-hand side of that equation then becomes

$$\frac{te^t}{e^t - 1} = \frac{-t}{e^{-t} - 1}. \quad (12.46)$$

Thus, equating Eq. (12.42) for $s = 1$ with the Bernoulli-number expansion (in $-t$) of the right-hand side of Eq. (12.46), we reach

$$\sum_{n=0}^{\infty} B_n(1) \frac{t^n}{n!} = \sum_{n=0}^{\infty} B_n \frac{(-t)^n}{n!},$$

which is equivalent to

$$B_n(1) = (-1)^n B_n(0). \quad (12.47)$$

These relations are used in the development of the Euler-Maclaurin integration formula.

Exercises

12.2.1 Verify the identities, Eqs. (12.32) and (12.46).

12.2.2 Show that the first Bernoulli polynomials are

$$B_0(s) = 1$$

$$B_1(s) = s - \frac{1}{2}$$

$$B_2(s) = s^2 - s + \frac{1}{6}.$$

Note that $B_n(0) = B_n$, the Bernoulli number.

12.2.3 Show that

$$\tan x = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 2^{2n} (2^{2n} - 1) B_{2n}}{(2n)!} x^{2n-1}, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Hint. $\tan x = \cot x - 2 \cot 2x$.

12.3 EULER-MACLAURIN INTEGRATION FORMULA

One use of the Bernoulli polynomials is in the derivation of the **Euler-Maclaurin integration formula**. This formula is used both to develop asymptotic expansions (treated later in this chapter) and to obtain approximate values for summations. An important application of the Euler-Maclaurin formula, presented in Chapter 13, is its use to derive **Stirling's formula**, an asymptotic expression for the gamma function.

The technique we use to develop the Euler-Maclaurin formula is repeated integration by parts, using Eq. (12.45) to create new derivatives. We start with

$$\int_0^1 f(x) dx = \int_0^1 f(x) B_0(x) dx, \quad (12.48)$$

where we have, for reasons that will shortly become apparent, inserted the redundant factor $B_0(x) = 1$. From Eq. (12.45), we note that

$$B_0(x) = B_1'(x),$$

and we substitute $B_1'(x)$ for $B_0(x)$ in Eq. (12.48), integrate by parts, and identify $B_1(1) = -B_1(0) = \frac{1}{2}$, thereby obtaining

$$\begin{aligned} \int_0^1 f(x) dx &= f(1)B_1(1) - f(0)B_1(0) - \int_0^1 f'(x)B_1(x) dx \\ &= \frac{1}{2}[f(1) + f(0)] - \int_0^1 f'(x)B_1(x) dx. \end{aligned} \quad (12.49)$$

Again using Eq. (12.45), we have

$$B_1(x) = \frac{1}{2}B_2'(x).$$

Inserting $B_2'(x)$ and integrating by parts again, we get

$$\begin{aligned} \int_0^1 f(x) dx &= \frac{1}{2}[f(1) + f(0)] - \frac{1}{2}[f'(1)B_2(1) - f'(0)B_2(0)] \\ &\quad + \frac{1}{2} \int_0^1 f^{(2)}(x)B_2(x) dx. \end{aligned} \quad (12.50)$$

Using the relation

$$B_{2n}(1) = B_{2n}(0) = B_{2n}, \quad n = 0, 1, 2, \dots, \quad (12.51)$$

Eq. (12.50) simplifies to

$$\int_0^1 f(x)dx = \frac{1}{2}[f(1) + f(0)] - \frac{B_2}{2}[f'(1) - f'(0)] + \frac{1}{2} \int_0^1 f^{(2)}(x)B_2(x)dx. \quad (12.52)$$

Continuing, we replace $B_2(x)$ by $B'_3(x)/3$ and once again integrate by parts. Because

$$B_{2n+1}(1) = B_{2n+1}(0) = 0, \quad n = 1, 2, 3, \dots, \quad (12.53)$$

the integration by parts produces no integrated terms, and

$$\frac{1}{2} \int_0^1 f^{(2)}(x)B_2(x)dx = \frac{1}{2 \cdot 3} \int_0^1 f^{(2)}(x)B'_3(x)dx = -\frac{1}{3!} \int_0^1 f^{(3)}(x)B_3(x)dx. \quad (12.54)$$

Substituting $B_3(x) = B'_4(x)/4$ and carrying out one more partial integration, we get integrated terms containing $B_4(x)$, which simplify according to Eq. (12.51). The result is

$$-\frac{1}{3!} \int_0^1 f^{(3)}(x)B_3(x)dx = \frac{B_4}{4!}[f^{(3)}(1) - f^{(3)}(0)] + \frac{1}{4!} \int_0^1 f^{(4)}(x)B_4(x)dx. \quad (12.55)$$

We may continue this process, with steps that are entirely analogous to those that led to Eqs. (12.54) and (12.55). After steps leading to derivatives of f of order $2q - 1$, we have

$$\begin{aligned} \int_0^1 f(x)dx &= \frac{1}{2}[f(1) + f(0)] - \sum_{p=1}^q \frac{1}{(2p)!} B_{2p}[f^{(2p-1)}(1) - f^{(2p-1)}(0)] \\ &\quad + \frac{1}{(2q)!} \int_0^1 f^{(2q)}(x)B_{2q}(x)dx. \end{aligned} \quad (12.56)$$

This is the Euler-Maclaurin integration formula. It assumes that the function $f(x)$ has the required derivatives.

The range of integration in Eq. (12.56) may be shifted from $[0, 1]$ to $[1, 2]$ by replacing $f(x)$ by $f(x + 1)$. Adding such results up to $[n - 1, n]$, we obtain

$$\begin{aligned} \int_0^n f(x) dx &= \frac{1}{2} f(0) + f(1) + f(2) + \cdots + f(n-1) + \frac{1}{2} f(n) \\ &\quad - \sum_{p=1}^q \frac{1}{(2p)!} B_{2p} \left[f^{(2p-1)}(n) - f^{(2p-1)}(0) \right] \\ &\quad + \frac{1}{(2q)!} \int_0^1 B_{2q}(x) \sum_{v=0}^{n-1} f^{(2q)}(x+v) dx. \end{aligned} \quad (12.57)$$

Note that the derivative terms at the intermediate integer arguments all cancel. However, the intermediate terms $f(j)$ do not, and $\frac{1}{2}f(0) + f(1) + \cdots + \frac{1}{2}f(n)$ appear exactly as in trapezoidal integration, or quadrature, so the summation over p may be interpreted as a correction to the trapezoidal approximation. Equation (12.57) may therefore be seen as a generalization of Eq. (1.10).

In many applications of Eq. (12.57) the final integral containing $f^{(2q)}$, though small, will not approach zero as q is increased without limit, and the Euler-Maclaurin formula then has an asymptotic, rather than convergent character. Such series, and the implications regarding their use, are the topic of a later section of this chapter.

One of the most important uses of the Euler-Maclaurin formula is in summing series by converting them to integrals plus correction terms.² Here is an illustration of the process.

Example 12.3.1 ESTIMATION OF $\zeta(3)$

A straightforward application of Eq. (12.57) to $\zeta(3)$ proceeds as follows (noting that all derivatives of $f(x) = 1/x^3$ vanish in the limit $x \rightarrow \infty$):

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{1}{2} f(1) + \int_1^{\infty} \frac{dx}{x^3} - \sum_{p=1}^q \frac{B_{2p}}{(2p)!} f^{(2p-1)}(1) + \text{remainder}. \quad (12.58)$$

Evaluating the integral, setting $f(1) = 1$, and inserting

$$f^{(2n-1)}(x) = -\frac{(2n+1)!}{2x^{2n+2}}$$

with $x = 1$, Eq. (12.58) becomes

$$\zeta(3) = \frac{1}{2} + \frac{1}{2} + \sum_{p=1}^q \frac{(2p+1)B_{2p}}{2x^{2p+2}} + \text{remainder}. \quad (12.59)$$

²See R. P. Boas and C. Stutz, Estimating sums with integrals. *Am. J. Phys.* **39**: 745 (1971), for a number of examples.

Table 12.4 Contributions to $\zeta(3)$ of Terms in Euler-Maclaurin Formula

	$n_0 = 1$	$n_0 = 2$	$n_0 = 4$
Explicit terms	0.500000	1.062500	1.169849
$\int_{n_0}^{\infty} x^{-3} dx$	0.500000	0.125000	0.031250
B_2 term	0.250000	0.015615	0.000977
B_4 term	-0.083333	-0.001302	-0.000020
B_6 term	0.083333	0.000326	0.000001
B_8 term	-0.150000	-0.000146	-0.000000
B_{10} term	0.416667	0.000102	0.000000
B_{12} term	-1.645238	-0.000100	-0.000000
B_{14} term	8.750000	0.000134	0.000000
Sum ^a	1.166667	1.201995	1.202057

^a Sums only include data above horizontal marker.

Left column: formula applied to entire summation; central column: formula applied starting from second term; right column: formula starting from fourth term.

To assess the quality of this result, we list, in the first data column of [Table 12.4](#), the contributions to it. The line marked “explicit terms” consists presently of only the term $\frac{1}{2}f(1)$. We note that the individual terms start to increase after the B_4 term; since it is our intention not to evaluate the remainder, the accuracy of the expansion is limited. As discussed more extensively in the section on asymptotic expansions, the best result available from these data is obtained by truncating the expansion before the terms start to increase; adding the contributions above the marker line in the table, we get the value listed as “Sum.” For reference, the accurate value of $\zeta(3)$ is 1.202057.

We can improve the result available from the Euler-Maclaurin formula by explicitly calculating some initial terms and applying the formula only to those that remain. This stratagem causes the derivatives entering the formula to be smaller and diminishes the correction from the trapezoid-rule estimate. Simply starting the formula at $n = 2$ instead of $n = 1$ reduces the error markedly; see the second data column of [Table 12.4](#). Now the “explicit terms” consist of $f(1) + \frac{1}{2}f(2)$. Starting the Euler-Maclaurin formula at $n = 4$ further improves the result, then reaching better than seven-figure accuracy. ■

When the Euler-Maclaurin formula is applied to sums whose summands have a finite number of nonzero derivatives, it can evaluate them exactly. See [Exercise 12.3.1](#).

Exercises

12.3.1 The Euler-Maclaurin integration formula may be used for the evaluation of finite series:

$$\sum_{m=1}^n f(m) = \int_1^n f(x) dx + \frac{1}{2}f(1) + \frac{1}{2}f(n) + \frac{B_2}{2!} [f'(n) - f'(1)] + \cdots.$$

Show that

- (a) $\sum_{m=1}^n m = \frac{1}{2}n(n+1).$
- (b) $\sum_{m=1}^n m^2 = \frac{1}{6}n(n+1)(2n+1).$
- (c) $\sum_{m=1}^n m^3 = \frac{1}{4}n^2(n+1)^2.$
- (d) $\sum_{m=1}^n m^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2+3n-1).$

12.3.2 The Euler-Maclaurin integration formula provides a way of calculating the Euler-Mascheroni constant γ to high accuracy. Using $f(x) = 1/x$ in Eq. (12.57) (with interval $[1, n]$) and the definition of γ , Eq. (1.13), we obtain

$$\gamma = \sum_{s=1}^n s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^N \frac{B_{2k}}{(2k)n^{2k}}.$$

Using double-precision arithmetic, calculate γ for $N = 1, 2, \dots$

Note. See D. E. Knuth, Euler's constant to 1271 places. *Math. Comput.* **16**: 275 (1962).

ANS. For $n = 1000$, $N = 2$
 $\gamma = 0.5772\ 1566\ 4901.$

12.4 DIRICHLET SERIES

Series expansions of the general form

$$S(s) = \sum_n \frac{a_n}{n^s}$$

are known as **Dirichlet series**, and our knowledge of contour integration methods and Bernoulli numbers enables us to evaluate a variety of expressions of this type. One of the most important Dirichlet series is that of the Riemann zeta function,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}. \quad (12.60)$$

We have already evaluated a sum from which $\zeta(2)$ can be extracted.

Example 12.4.1 EVALUATION OF $\zeta(2)$

From Example 11.9.1, we have

$$S(a) = \sum_{n=1}^{\infty} \frac{1}{n^2 + a^2} = \frac{\pi \coth \pi a}{2a} - \frac{1}{2a^2}.$$

Simply by taking the limit $a \rightarrow 0$, we have

$$\zeta(2) = \lim_{a \rightarrow 0} S(a) = \lim_{a \rightarrow 0} \left[\frac{\pi}{2a} \left(\frac{1}{\pi a} + \frac{\pi a}{3} + \cdots \right) - \frac{1}{2a^2} \right] = \frac{\pi^2}{6}. \quad (12.61)$$

■

From the relation with the Bernoulli numbers, or alternatively (and perhaps less conveniently) by contour-integration methods, we find

$$\zeta(4) = \frac{\pi^4}{90}.$$

Values of $\zeta(2n)$ through $\zeta(10)$ are listed in [Exercise 12.4.1](#). The zeta functions of odd integer argument seem unamenable to evaluation in closed form, but are easy to compute numerically (see [Example 12.3.1](#)).

Other useful Dirichlet series, in the notation of AMS-55 (see Additional Readings), include

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} = (1 - 2^{1-s})\zeta(s), \quad (12.62)$$

$$\lambda(s) = \sum_{n=0}^{\infty} (2n+1)^{-s} = (1 - 2^{-s})\zeta(s), \quad (12.63)$$

$$\beta(s) = \sum_{n=0}^{\infty} (-1)^n (2n+1)^{-s}. \quad (12.64)$$

Closed expressions are available (for integer $n \geq 1$) for $\zeta(2n)$, $\eta(2n)$, and $\lambda(2n)$, and for $\beta(2n-1)$. The sums with exponents of opposite parity cannot be reduced to $\zeta(2n)$ or performed by the contour-integral methods we discussed in Chapter 11. An important series that can only be evaluated numerically is that whose result is **Catalan's constant**, which is

$$\beta(2) = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \cdots = 0.91596559 \dots \quad (12.65)$$

For reference, we list a few of these summable Dirichlet series:

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}, \quad (12.66)$$

$$\zeta(4) = 1 + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^4}{90}, \quad (12.67)$$

$$\eta(2) = 1 - \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{12}, \quad (12.68)$$

$$\eta(4) = 1 - \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{7\pi^4}{720}, \quad (12.69)$$

$$\lambda(2) = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}, \quad (12.70)$$

$$\lambda(4) = 1 + \frac{1}{3^4} + \frac{1}{5^4} + \cdots = \frac{\pi^4}{96}, \quad (12.71)$$

$$\beta(1) = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}, \quad (12.72)$$

$$\beta(3) = 1 - \frac{1}{3^3} + \frac{1}{5^3} - \cdots = \frac{\pi^3}{32}. \quad (12.73)$$

Exercises

12.4.1 From $B_{2n} = (-1)^{n-1} \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n)$, show that

$$\begin{aligned} \text{(a) } \zeta(2) &= \frac{\pi^2}{6}, & \text{(d) } \zeta(8) &= \frac{\pi^8}{9450}, \\ \text{(b) } \zeta(4) &= \frac{\pi^4}{90}, & \text{(e) } \zeta(10) &= \frac{\pi^{10}}{93,555}. \\ \text{(c) } \zeta(6) &= \frac{\pi^6}{945}, \end{aligned}$$

12.4.2 The integral

$$\int_0^1 [\ln(1-x)]^2 \frac{dx}{x}$$

appears in the fourth-order correction to the magnetic moment of the electron. Show that it equals $2\zeta(3)$.

Hint. Let $1-x = e^{-t}$.

12.4.3 (a) Show that

$$\int_0^{\infty} \frac{(\ln z)^2}{1+z^2} dz = 4 \left(1 - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots \right).$$

(b) By contour integration show that this series evaluates to $\pi^3/8$.

12.4.4 Show that Catalan's constant, $\beta(2)$, may be written as

$$\beta(2) = 2 \sum_{k=1}^{\infty} (4k-3)^{-2} - \frac{\pi^2}{8}.$$

Hint. $\pi^2 = 6\zeta(2)$.

12.4.5 Show that

$$(a) \int_0^1 \frac{\ln(1+x)}{x} dx = \frac{1}{2}\zeta(2), \quad (b) \lim_{a \rightarrow 1} \int_0^a \frac{\ln(1-x)}{x} dx = \zeta(2).$$

Note that the integrand in part (b) diverges for $a = 1$ but that the integral is convergent.

12.4.6 (a) Show that the equation $\ln 2 = \sum_{s=1}^{\infty} (-1)^{s+1} s^{-1}$, Eq. (1.53), may be rewritten as

$$\ln 2 = \sum_{s=2}^n 2^{-s} \zeta(s) + \sum_{p=1}^{\infty} (2p)^{-n-1} \left[1 - \frac{1}{2p} \right]^{-1}.$$

Hint. Take the terms in pairs.

(b) Calculate $\ln 2$ to six significant figures.

12.4.7 (a) Show that the equation $\pi/4 = \sum_{s=1}^{\infty} (-1)^{s+1} (2s-1)^{-1}$, Eq. (12.72), may be rewritten as

$$\frac{\pi}{4} = 1 - 2 \sum_{s=1}^n 4^{-2s} \zeta(2s) - 2 \sum_{p=1}^{\infty} (4p)^{-2n-2} \left[1 - \frac{1}{(4p)^2} \right]^{-1}.$$

(b) Calculate $\pi/4$ to six significant figures.

12.5 INFINITE PRODUCTS

We saw in Chapter 11 that complex variable theory can be used to generate infinite-product representations of analytic functions. Here we develop some of their properties. For that purpose it is convenient to write these products in the form

$$P = \prod_{n=1}^{\infty} (1 + a_n).$$

The infinite product may be related to an infinite series by the obvious method of taking the logarithm:

$$\ln \prod_{n=1}^{\infty} (1 + a_n) = \sum_{n=1}^{\infty} \ln(1 + a_n). \quad (12.74)$$

The main theorem regarding convergence of infinite products is the following:

If $0 \leq a_n < 1$, the infinite products $\prod_{n=1}^{\infty} (1 + a_n)$ and $\prod_{n=1}^{\infty} (1 - a_n)$ converge if $\sum_{n=1}^{\infty} a_n$ converges and diverge if $\sum_{n=1}^{\infty} a_n$ diverges.

For the infinite product $\prod (1 + a_n)$, note that

$$1 + a_n \leq e^{a_n},$$

which means that the partial product consisting of the first n factors satisfies

$$p_n \leq e^{s_n},$$

where s_n is the sum of the first n a_n . Letting $n \rightarrow \infty$,

$$\prod_{n=1}^{\infty} (1 + a_n) \leq \exp \sum_{n=1}^{\infty} a_n, \quad (12.75)$$

thereby giving an upper bound for the infinite product.

To develop a lower bound, we note that, because all $a_i > 0$,

$$p_n = 1 + \sum_{i=1}^n a_i + \sum_{i=1}^n \sum_{j=1}^n a_i a_j + \cdots \geq s_n.$$

Hence

$$\prod_{n=1}^{\infty} (1 + a_n) \geq \sum_{n=1}^{\infty} a_n. \quad (12.76)$$

If the infinite sum remains finite, the infinite product will also. But if the infinite sum diverges, so will the infinite product.

The case $\prod (1 - a_n)$ is complicated by the negative signs, but a proof similar to the foregoing may be developed by noting that for $a_n < \frac{1}{2}$,

$$(1 - a_n) \leq (1 + a_n)^{-1} \quad \text{and} \quad (1 - a_n) \geq (1 + 2a_n)^{-1}.$$

Example 12.5.1 CONVERGENCE OF INFINITE PRODUCTS FOR $\sin z$ AND $\cos z$

These products, developed in Eqs. (11.89) and (11.90), are

$$\sin z = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2 \pi^2} \right), \quad \cos z = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{(n - 1/2)^2 \pi^2} \right). \quad (12.77)$$

The product expansion of $\sin z$ converges for all z , because, writing the factors as $(1 - a_n)$,

$$\sum_{n=1}^{\infty} a_n = \frac{z^2}{\pi^2} \sum_{n=1}^{\infty} n^{-2} = \frac{z^2}{\pi^2} \zeta(2) = \frac{z^2}{6},$$

a convergent result. For the expansion of $\cos z$, we have

$$\sum_{n=1}^{\infty} a_n = \frac{4z^2}{\pi^2} \sum_{n=1}^{\infty} (2n-1)^{-2} = \frac{4z^2}{\pi^2} \lambda(2) = \frac{z^2}{2},$$

also convergent for all z . Note, however, that if z is large, many terms of the product will have to be taken before either of these series approaches convergence. In fact, the main use of these series is in establishing mathematical results rather than for precise numerical work in physics. ■

We close this section with one further example illustrating a technique for working with infinite products.

Example 12.5.2 AN INTERESTING PRODUCT

We wish to evaluate the infinite product

$$P = \prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2}\right).$$

We note that the product we seek is equivalent to all but the first term of the product expansion of $\sin z$ with $z = \pi$ as given in Eq. (12.77). In fact, the missing first term, which is zero, guarantees that we will get the correct result for $\sin \pi$. For general z , we move the first term (and the prefactor z) to the left-hand side of the product formula for $\sin z$, reaching

$$\frac{\sin z}{z(1 - z^2/\pi^2)} = \prod_{n=2}^{\infty} \left(1 - \frac{z^2}{n^2\pi^2}\right).$$

We now take the limits of the two sides of this equation as $z \rightarrow \pi$, applying l'Hôpital's rule to evaluate the left-hand side and recognizing the right-hand side as P . Thus,

$$P = \lim_{z \rightarrow \pi} \frac{\sin z}{z(1 - z^2/\pi^2)} = \frac{\cos z}{1 - 3z^2/\pi^2} \Big|_{z=\pi} = \frac{-1}{1 - 3} = +\frac{1}{2}.$$

■

Exercises

12.5.1 Using

$$\ln \prod_{n=1}^{\infty} (1 \pm a_n) = \sum_{n=1}^{\infty} \ln(1 \pm a_n)$$

and the Maclaurin expansion of $\ln(1 \pm a_n)$, show that the infinite product $\prod_{n=1}^{\infty} (1 \pm a_n)$ converges or diverges with the infinite series $\sum_{n=1}^{\infty} a_n$.

12.5.2 An infinite product appears in the form

$$\prod_{n=1}^{\infty} \left(\frac{1 + a/n}{1 + b/n} \right),$$

where a and b are constants. Show that this infinite product converges only if $a = b$.

12.5.3 Show that the infinite product representations of $\sin x$ and $\cos x$ are consistent with the identity $2 \sin x \cos x = \sin 2x$.

12.5.4 Determine the limit to which $\prod_{n=2}^{\infty} \left(1 + \frac{(-1)^n}{n} \right)$ converges.

12.5.5 Show that $\prod_{n=2}^{\infty} \left[1 - \frac{2}{n(n+1)} \right] = \frac{1}{3}$.

12.5.6 Prove that $\prod_{n=2}^{\infty} \left(1 - \frac{1}{n^2} \right) = \frac{1}{2}$.

12.5.7 Verify the Euler identity $\prod_{p=1}^{\infty} (1 + z^p) = \prod_{q=1}^{\infty} (1 - z^{2q-1})^{-1}$, $|z| < 1$.

12.5.8 Show that $\prod_{r=1}^{\infty} (1 + x/r) e^{-x/r}$ converges for all finite x (except for the zeros of $1 + x/r$).

Hint. Write the n th factor as $1 + a_n$.

12.5.9 Derive the formula, valid for small x ,

$$\ln \sin x = \ln x + \sum a_n x^n,$$

giving the explicit form for the coefficients a_n .

Hint. $d(\ln \sin x)/dx = \cot x$.

12.5.10 Using the infinite product representations of $\sin z$, show that

$$z \cot z = 1 - 2 \sum_{m,n=1}^{\infty} \left(\frac{z}{n\pi} \right)^{2m},$$

and hence that the Bernoulli numbers are given by the formula

$$B_{2n} = (-1)^{n-1} \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n).$$

This is an alternate route to [Eq. \(12.38\)](#).

Hint. The result of [Exercise 12.5.9](#) will be helpful.

12.6 ASYMPTOTIC SERIES

Asymptotic series frequently occur in physics. In fact, one of the earliest and still important approximations of quantum mechanics, the **WKB expansion** (the initials stand for its originators, Wenzel, Kramers, and Brillouin), is an asymptotic series. In numerical computations, these series are employed for the accurate computation of a variety of functions.

We consider here two types of integrals that lead to asymptotic series: first, integrals of the form

$$I_1(x) = \int_x^{\infty} e^{-u} f(u) du,$$

where the variable x appears as the lower limit of an integral. Second, we consider the form

$$I_2(x) = \int_0^{\infty} e^{-u} f\left(\frac{u}{x}\right) du,$$

with the function f to be expanded as a Taylor series (binomial series). Asymptotic series often occur as solutions of differential equations; we encounter many examples in later chapters of this book.

Exponential Integral

The nature of an asymptotic series is perhaps best illustrated by a specific example. Suppose that we have the exponential integral function³

$$\text{Ei}(x) = \int_{-\infty}^x \frac{e^u}{u} du, \quad (12.78)$$

which we find more convenient to write in the form

$$-\text{Ei}(-x) = \int_x^{\infty} \frac{e^{-u}}{u} du = E_1(x), \quad (12.79)$$

to be evaluated for large values of x . This function has a series expansion that converges for all x , namely

$$E_1(x) = -\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{nn!}, \quad (12.80)$$

which we derive in Chapter 13, but the series is totally useless for numerical evaluation when x is large. We need another approach, for which it is convenient to generalize Eq. (12.79) to

$$I(x, p) = \int_x^{\infty} \frac{e^{-u}}{u^p} du, \quad (12.81)$$

where we restrict consideration to cases in which x and p are positive. As already stated, we seek an evaluation for large values of x .

³This function occurs frequently in astrophysical problems involving gas with a Maxwell-Boltzmann energy distribution.

Integrating by parts, we obtain

$$I(x, p) = \frac{e^{-x}}{x^p} - p \int_x^\infty \frac{e^{-u}}{u^{p+1}} du = \frac{e^{-x}}{x^p} - \frac{pe^{-x}}{x^{p+1}} + p(p+1) \int_x^\infty \frac{e^{-u}}{u^{p+2}} du.$$

Continuing to integrate by parts, we develop the series

$$\begin{aligned} I(x, p) = e^{-x} \left(\frac{1}{x^p} - \frac{p}{x^{p+1}} + \frac{p(p+1)}{x^{p+2}} - \dots + (-1)^{n-1} \frac{(p+n-2)!}{(p-1)!x^{p+n-1}} \right) \\ + (-1)^n \frac{(p+n-1)!}{(p-1)!} \int_x^\infty \frac{e^{-u}}{u^{p+n}} du. \end{aligned} \quad (12.82)$$

This is a remarkable series. Checking the convergence by the d'Alembert ratio test, we find

$$\lim_{n \rightarrow \infty} \frac{|u_{n+1}|}{|u_n|} = \lim_{n \rightarrow \infty} \frac{(p+n)!}{(p+n-1)!} \cdot \frac{1}{x} = \lim_{n \rightarrow \infty} \frac{p+n}{x} = \infty \quad (12.83)$$

for all finite values of x . Therefore our series as an infinite series diverges everywhere! Before discarding Eq. (12.83) as worthless, let us see how well a given partial sum approximates our function $I(x, p)$. Taking s_n as the partial sum of the series through n terms and R_n as the corresponding remainder,

$$I(x, p) - s_n(x, p) = (-1)^{n+1} \frac{(p+n)!}{(p-1)!} \int_x^\infty \frac{e^{-u}}{u^{p+n+1}} du = R_n(x, p).$$

In absolute value

$$|R_n(x, p)| \leq \frac{(p+n)!}{(p-1)!} \int_x^\infty \frac{e^{-u}}{u^{p+n+1}} du.$$

When we substitute $u = v + x$, the integral becomes

$$\begin{aligned} \int_x^\infty \frac{e^{-u}}{u^{p+n+1}} du &= e^{-x} \int_0^\infty \frac{e^{-v}}{(v+x)^{p+n+1}} dv \\ &= \frac{e^{-x}}{x^{p+n+1}} \int_0^\infty e^{-v} \left(1 + \frac{v}{x}\right)^{-p-n-1} dv. \end{aligned}$$

For large x the final integral approaches 1 and

$$|R_n(x, p)| \approx \frac{(p+n)!}{(p-1)!} \frac{e^{-x}}{x^{p+n+1}}. \quad (12.84)$$

This means that if we take x large enough, our partial sum s_n will be an arbitrarily good approximation to the function $I(x, p)$. Our divergent series, Eq. (12.82), therefore is perfectly good for computations of partial sums. For this reason it is sometimes called a **semi-convergent series**. Note that the power of x in the denominator of the remainder, namely $p + n + 1$, is higher than the power of x in the last term included in $s_n(x, p)$, namely $p + n$.

Thus, our asymptotic series for $E_1(x)$ assumes the form

$$e^x E_1(x) = e^x \int_x^\infty \frac{e^{-u}}{u} du$$

$$\approx s_n(x) = \frac{1}{x} - \frac{1!}{x^2} + \frac{2!}{x^2} - \frac{3!}{x^4} + \cdots + (-1)^n \frac{n!}{x^{n+1}}, \quad (12.85)$$

where we must choose to terminate the series after some n .

Since the remainder $R_n(x, p)$ alternates in sign, the successive partial sums give alternately upper and lower bounds for $I(x, p)$. The behavior of the series (with $p = 1$) as a function of the number of terms included is shown in Fig. 12.2, where we have plotted partial sums of $e^x E_1(x)$ for the value $x = 5$. The optimum determination of $e^x E_1(x)$ is given by the closest approach of the upper and lower bounds, that is, for $x = 5$, between $s_6 = 0.1664$ and $s_5 = 0.1741$. Therefore

$$0.1664 \leq e^x E_1(x) \Big|_{x=5} \leq 0.1741. \quad (12.86)$$

Actually, from tables,

$$e^x E_1(x) \Big|_{x=5} = 0.1704, \quad (12.87)$$

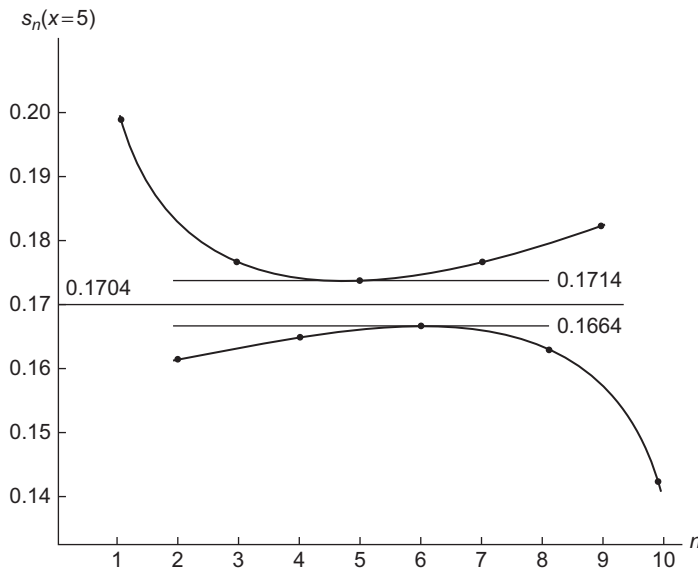


FIGURE 12.2 Partial sums of $e^x E_1(x) \Big|_{x=5}$.

within the limits established by our asymptotic expansion. Note that inclusion of additional terms in the series expansion beyond the optimum point reduces the accuracy of the representation. As x is increased, the spread between the lowest upper bound and the highest lower bound will diminish. By taking x large enough, one may compute $e^x E_1(x)$ to any desired degree of accuracy. Other properties of $E_1(x)$ are derived and discussed in Section 13.6.

Cosine and Sine Integrals

Asymptotic series may also be developed from definite integrals, provided that the integrand has the required behavior. As an example, the cosine and sine integrals (in Table 1.2) are defined by

$$\text{Ci}(u) = - \int_u^{\infty} \frac{\cos t}{t} dt, \quad (12.88)$$

$$\text{si}(u) = - \int_u^{\infty} \frac{\sin t}{t} dt. \quad (12.89)$$

Combining these, using the formula for e^{it} ,

$$\text{Ci}(u) + i \text{si}(u) = - \int_u^{\infty} \frac{e^{-it}}{t} dt,$$

and then changing the integration variable from t to z , we reach

$$F(u) = \text{Ci}(u) + i \text{si}(u) = -e^{iu} \int_0^{\infty} \frac{e^{iz} dz}{u+z}. \quad (12.90)$$

To further process $F(u)$, we now consider the contour integral

$$-e^{iu} \oint_C \frac{e^{iz} dz}{u+z},$$

where the contour C is that shown in Fig. 12.3. Since we are interested in evaluation for large positive (and real) u , our integrand has as its only singularity a pole on the negative real axis, so the region enclosed by the contour is entirely analytic and the contour integral therefore vanishes. The exponential and the denominator cause the arc at infinity (labeled B) not to contribute to the contour integral, so the integral we seek is obtained from segment A and must be equal to the negative of the integral on segment D . Therefore, we have

$$F(u) = -e^{iu} \int_0^{\infty} \frac{e^{-y} i dy}{u+iy}, \quad (12.91)$$

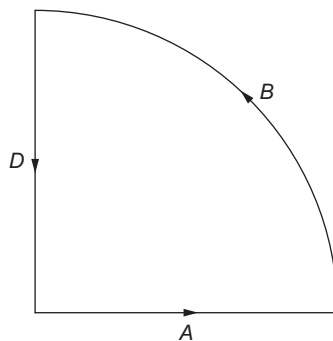


FIGURE 12.3 Contour for sine and cosine integrals.

which is already helpful since we have converted an oscillatory integral into one with a monotonically and exponentially decreasing integrand. To obtain an asymptotic expansion, we continue by expanding the denominator of the integrand using the binomial theorem, writing

$$\frac{1}{u + iy} = \frac{1}{u} \left[1 - \frac{iy}{u} + \left(\frac{iy}{u} \right)^2 - \cdots \right].$$

We plan to integrate in y from zero to infinity, and the proposed expansion will be divergent when $y > u$, but we proceed anyway, because the terms of the series will initially be decreasing and will be satisfactory as an asymptotic expansion. Formally, we take the viewpoint that we are writing $1/(u + iy)$ as a finite series plus a remainder, and we will abandon the expansion at or before the point that the remainder is a minimum.

Inserting the expansion, and integrating termwise using the formula

$$\int_0^\infty y^n e^{-y} dy = n!,$$

we get

$$F(u) \approx -\frac{ie^{iu}}{u} \left[1 - i \left(\frac{1!}{u} \right) - \left(\frac{2!}{u^2} \right) + i \left(\frac{3!}{u^3} \right) + \left(\frac{4!}{u^4} \right) - \cdots \right]. \quad (12.92)$$

As for our earlier example, the exponential integral, this series will diverge for all u , but if u is sufficiently large the terms will initially decrease to very small values before increasing again toward divergence.

To go from the expansion of $F(u)$ to those of Ci and si, we need to separate it into real and imaginary parts. Writing $e^{iu} = \cos u + i \sin u$ and collecting terms appropriately, we

get as the desired asymptotic expansions

$$\text{Ci}(u) \approx \frac{\sin u}{u} \sum_{n=0}^N (-1)^n \frac{(2n)!}{u^{2n}} - \frac{\cos u}{u} \sum_{n=0}^N (-1)^n \frac{(2n+1)!}{u^{2n+1}}, \quad (12.93)$$

$$\text{si}(u) \approx -\frac{\cos u}{u} \sum_{n=0}^N (-1)^n \frac{(2n)!}{u^{2n}} - \frac{\sin u}{u} \sum_{n=0}^N (-1)^n \frac{(2n+1)!}{u^{2n+1}}. \quad (12.94)$$

Definition of Asymptotic Series

Poincaré has introduced a formal definition for an asymptotic series.⁴ Following Poincaré, we consider a function $f(x)$ whose asymptotic expansion is sought, the partial sums s_n in its expansion, and the corresponding remainders $R_n(x)$. Though the expansion need not be a power series, we assume that form for simplicity in the present discussion. Thus,

$$x^n R_n(x) = x^n [f(x) - s_n(x)], \quad (12.95)$$

where

$$s_n(x) = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}. \quad (12.96)$$

The asymptotic expansion of $f(x)$ is defined to have the properties that

$$\lim_{x \rightarrow \infty} x^n R_n(x) = 0, \text{ for fixed } n, \quad (12.97)$$

and

$$\lim_{n \rightarrow \infty} x^n R_n(x) = \infty, \text{ for fixed } x. \quad (12.98)$$

These conditions were met for our examples, Eqs. (12.85), (12.93), and (12.94).⁵

For power series, as assumed in the form of $s_n(x)$, $R_n(x) \approx x^{-n-1}$. With the conditions of Eqs. (12.97) and (12.98) satisfied, we write

$$f(x) \sim \sum_{n=0}^{\infty} a_n x^{-n}. \quad (12.99)$$

Note the use of $\sim$ in place of $=$. The function $f(x)$ is equal to the series only in the limit as $x \rightarrow \infty$ and with the restriction to a finite number of terms in the series.

Asymptotic expansions of two functions may be multiplied together, and the result will be an asymptotic expansion of the product of the two functions. The asymptotic expansion of a given function $f(t)$ may be integrated term by term (just as in a uniformly convergent series of continuous functions) from $x \leq t < \infty$, and the result will be an asymptotic

⁴Poincaré's definition allows (or neglects) exponentially decreasing functions. The refinement of his definition is of considerable importance for the advanced theory of asymptotic expansions, particularly for extensions into the complex plane. However, for purposes of an introductory treatment and especially for numerical computation of expansions for which the variable is real and positive, Poincaré's approach is perfectly satisfactory.

⁵Some writers feel that the requirement of Eq. (12.98), which excludes convergent series of inverse powers of x , is artificial and unnecessary.

expansion of $\int_x^\infty f(t)dt$. Term-by-term differentiation, however, is valid only under very special conditions.

Some functions do not possess an asymptotic expansion; e^x is an example of such a function. However, if a function has an asymptotic expansion of the power-series form in Eq. (12.99), it has only one. The correspondence is not one to one; many functions may have the same asymptotic expansion.

One of the most useful and powerful methods of generating asymptotic expansions, the method of steepest descents, is developed in the next section of this text.

Exercises

- 12.6.1** Integrating by parts, develop asymptotic expansions of the Fresnel integrals

$$(a) C(x) = \int_0^x \cos \frac{\pi u^2}{2} du, \quad (b) s(x) = \int_0^x \sin \frac{\pi u^2}{2} du.$$

These integrals appear in the analysis of a knife-edge diffraction pattern.

- 12.6.2** Rederive the asymptotic expansions of $\text{Ci}(x)$ and $\text{si}(x)$ by repeated integration by parts.

$$\text{Hint. } \text{Ci}(x) + i \text{si}(x) = - \int_x^\infty \frac{e^{it}}{t} dt.$$

- 12.6.3** Derive the asymptotic expansion of the Gauss error function

$$\begin{aligned} \text{erf}(x) &= \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \\ &\approx 1 - \frac{e^{-x^2}}{\sqrt{\pi}x} \left(1 - \frac{1}{2x^2} + \frac{1 \cdot 3}{2^2 x^4} - \frac{1 \cdot 3 \cdot 5}{2^3 x^6} + \cdots + (-1)^n \frac{(2n-1)!!}{2^n x^{2n}} \right). \end{aligned}$$

$$\text{Hint. } \text{erf}(x) = 1 - \text{erfc}(x) = 1 - \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt.$$

Normalized so that $\text{erf}(\infty) = 1$, this function plays an important role in probability theory. It may be expressed in terms of the Fresnel integrals (Exercise 12.6.1), the incomplete gamma functions (Section 13.6), or the confluent hypergeometric functions (Section 18.5).

- 12.6.4** The asymptotic expressions for the various Bessel functions, Section 14.6, contain the series

$$\begin{aligned} P_\nu(z) &\sim 1 + \sum_{n=1}^{\infty} (-1)^n \frac{\prod_{s=1}^{2n} [4\nu^2 - (2s-1)^2]}{(2n)!(8z)^{2n}}, \\ Q_\nu(z) &\sim \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\prod_{s=1}^{2n-1} [4\nu^2 - (2s-1)^2]}{(2n-1)!(8z)^{2n-1}}. \end{aligned}$$

Show that these two series are indeed asymptotic series.

12.6.5 For $x > 1$,

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{x^{n+1}}.$$

Test this series to see if it is an asymptotic series.

12.6.6 Derive the following Bernoulli-number asymptotic series for the Euler-Mascheroni constant, defined in Eq. (1.13):

$$\gamma \sim \sum_{s=1}^n s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)n^{2k}}.$$

Here n plays the role of x .

Hint. Apply the Euler-Maclaurin integration formula to $f(x) = x^{-1}$ over the interval $[1, n]$ for $N = 1, 2, \dots$

12.6.7 Develop an asymptotic series for

$$\int_0^{\infty} \frac{e^{-xv}}{(1+v^2)^2} dv.$$

Take x to be real and positive.

$$ANS. \quad \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \dots + \frac{(-1)^n (2n)!}{x^{2n+1}}.$$

12.7 METHOD OF STEEPEST DESCENTS

In this section we consider the frequently occurring situation that we require the asymptotic behavior (for large t , assumed real) of a function $f(t)$, where

- $f(t)$ is represented by an integral of the generic form

$$f(t) = \int_C F(z, t) dz,$$

with $F(z, t)$ analytic in z , but also parametrically dependent on t ;

- The integration path C is, or can be deformed to be, such that for large t the dominant contribution to the integral arises from a small range of z in the neighborhood of the point z_0 where $|F(z_0, t)|$ is a maximum **on the path**;
- The integration path will pass through z_0 in the orientation that causes the most rapid decrease in $|F|$ on departure from z_0 in either direction along the path (hence the name **steepest descents**); and
- In the limit of large t the contribution to the integral from the neighborhood of z_0 asymptotically approaches the exact value of $f(t)$.

While the above conditions seem rather restrictive, they can in fact be met for many of the important special functions of mathematical physics, including, among others, the gamma function and various Bessel functions.

Saddle Points

The integration path supplied with the original definition of an integral representation defining a function $f(t)$ will not usually meet the conditions outlined above, and we need to consider the features of the integrand $F(z, t)$ that will be useful in defining a more suitable path which, even if the original formulation is entirely real, may be a more general contour in the complex plane. We already know (Exercise 11.2.2) that neither the real nor the imaginary part of an analytic function can have an extremum (either a minimum or maximum) within the region of analyticity, and the same is also true of its modulus (this result is Jensen's theorem; see Exercise 12.7.1). To better understand that, let us represent $F(z, t)$ (in a region where it is assumed nonzero) in the form

$$F(z, t) = e^{w(z, t)} = e^{u(z, t) + iv(z, t)}, \quad (12.100)$$

where u and v are the real and imaginary parts of an analytic function w ; this representation permits us to identify u as $\ln |F|$; the fact that u cannot have an extremum makes Jensen's theorem obvious.

Although u cannot have an extremum, it can have a saddle point (a point at which $w' = 0$; then also $du/ds = 0$ for all directions ds , but with higher derivatives that are positive in some directions and negative in others (see Fig. 12.4). Let us examine some general features of w and its components u and v in the neighborhood of a saddle point of u , which we designate z_0 . We proceed by expanding $w(z, t)$ in a Taylor series about z_0 . Because $w' = 0$ there, the first two nonzero terms of the expansion are

$$w(z, t) = w(z_0, t) + \frac{w''(z_0, t)}{2!}(z - z_0)^2 + \cdots \quad (12.101)$$

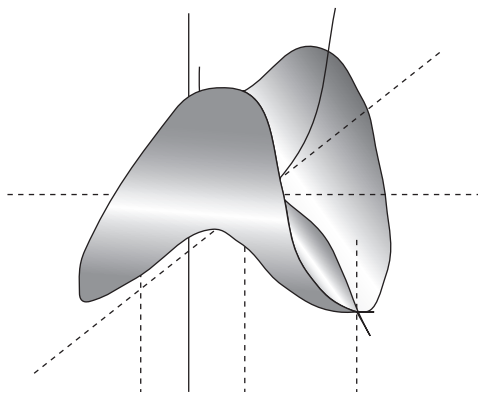


FIGURE 12.4 Saddle point of u ($= |F|$); see Eq. (12.100).

It could be that $w''(z_0, t) = 0$, but that possibility makes the analysis more complicated without changing it in a fundamental way, so we proceed under the assumption that $w''(z_0, t) \neq 0$. Using the abbreviated notations $w_0 = w(z_0, t)$, $w''(z_0, t) = w_0''$, and introducing the polar forms $w_0'' = |w_0''|e^{i\alpha}$, $z - z_0 = re^{i\theta}$, Eq. (12.101) becomes

$$w(z, t) = w_0 + \frac{1}{2}|w_0''|e^{i(\alpha+2\theta)}r^2 + \dots \quad (12.102)$$

$$= w_0 + \frac{1}{2}|w_0''|r^2 \left[\cos(\alpha + 2\theta) + i \sin(\alpha + 2\theta) \right] + \dots \quad (12.103)$$

For later reference we note that α is the **argument** of $w''(z_0, t)$. We see that, in general at a saddle point, u (the real part of w) will increase most rapidly when $\alpha + 2\theta = 2n\pi$, corresponding to the opposite directions $\theta = -\alpha/2$ and $\theta = -\alpha/2 + \pi$. On the other hand, u will decrease most rapidly when $\alpha + 2\theta = (2n+1)\pi$, i.e., $\theta = -\alpha/2 + (\frac{1}{2}\pi \text{ or } \frac{3}{2}\pi)$, the two directions perpendicular to those of maximum increase. And u will (to second order) remain constant (so-called **level lines**) in the directions $\theta = -\alpha/2 + (\frac{1}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi)$. See the left panel of Fig. 12.5.

The behavior of v (the imaginary part of w) will be similar to that of u , but displaced in angle by 45° . The level lines of v will be in the directions $\theta = -\alpha_s + (0, \pi/2, \pi, 3\pi/2)$, and therefore will coincide with the directions of maximum increase or decrease in u . See the right panel of Fig. 12.5.

We are now ready to identify an optimum contour for evaluating the integral representation of $f(t)$, namely one that passes through the saddle point z_0 in the directions of maximum rate of decrease in u with distance from z_0 , and therefore also in $|F|$. These directions have the additional advantage that they are level lines of v , so that the factor e^{iv} will not produce changes of phase (oscillatory behavior and therefore numerical instability) in F as we leave the saddle point. If we had chosen z_0 to be a point other than a saddle point, the expansion of w would have contained a nonzero linear term in r , and it would not have been possible to construct a curve through z_0 that would cause $|F|$ to decrease in both path directions, or to keep the phase of F constant.

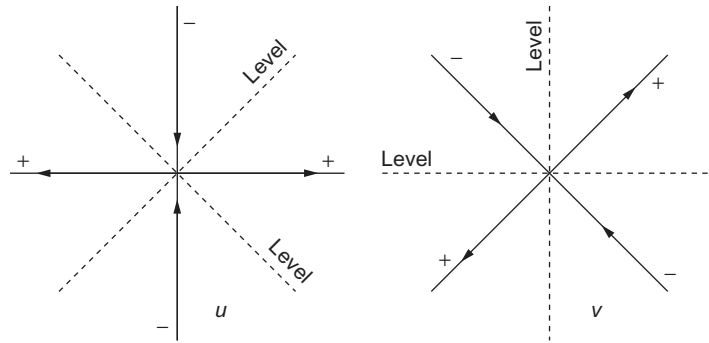


FIGURE 12.5 Near a saddle point in $w = u + iv$: When features of u are oriented as in the left panel, those of v are as shown in the right panel. Arrows indicate ascending directions.

Saddle Point Method

Now that we have identified z_0 and the directions of steepest descent in $|F(z, t)|$, we complete the specification of the method of steepest descents, also called the **saddle point method** of asymptotic approximation, by assuming that the significant contributions to the integral are from a small range of $0 \leq r \leq a$ in each of the two directions along the path. Before obtaining a final result, we must make one more observation. Looking at the way in which the contour had to be deformed to pass through z_0 , we need to determine the sense of the path (i.e., we must decide whether the direction of travel is at $\theta = -\alpha/2 + \frac{1}{2}\pi$ or at $\theta = -\alpha/2 + \frac{3}{2}\pi$). Assuming that this has been decided, we can then identify, for the portion of the path in which we descend from $F(z_0)$, $dz = e^{i\theta} dr$. The contribution in which we ascend to $F(z_0)$ will have the opposite sign for dz but we can handle it simply by multiplying the descending contribution by two. Then, noting that $e^{i(\alpha+2\theta)} = -1$, our approximation to $f(t)$ is

$$f(t) \approx 2e^{w_0+i\theta} \int_0^a e^{-|w_0''|r^2/2} dr, \quad (12.104)$$

where the initial “2” causes inclusion of the ascent to z_0 . We now make the key assumption of the method, namely that $|w_0''|$, the measure of the rate of decrease in $|F|$ as we leave z_0 , is large enough that the bulk of the value of the integral has already been attained for small a , and that the exponential decrease in the value of the integrand enables us to replace a by infinity without making significant error. In problems where the saddle point method is applicable, this condition is met when t is sufficiently large. We complete the present analysis by remembering that $e^{w_0} = F(z_0, t)$ and by evaluating the integral for $a = \infty$, where, cf. Eq. (1.148), it has the value $\sqrt{\pi/2|w_0''|}$. We get

$$f(t) \approx F(z_0, t) e^{i\theta} \sqrt{\frac{2\pi}{|w''(z_0, t)|}}. \quad (12.105)$$

We remind the reader that

$$\theta = -\frac{\arg(w''(z_0, t))}{2} + \left(\frac{\pi}{2} \text{ or } \frac{3\pi}{2} \right), \quad (12.106)$$

with the choice (which affects only the sign of the final result) determined from the sense in which the contour passes through the saddle point z_0 .

Sometimes it is sufficient to apply the method of steepest descents only to the rapidly varying part of an integral. This corresponds to assuming that we may make the approximation

$$f(t) = \int_C g(z, t) F(z, t) dz \approx g(z_0, t) \int_C F(z, t) dz, \quad (12.107)$$

after which we proceed as before. Note that this causes g not to be considered when we define w or w'' , and our final formula is replaced by

$$f(t) \approx g(z_0, t) F(z_0, t) e^{i\theta} \sqrt{\frac{2\pi}{|w''(z_0, t)|}}. \quad (12.108)$$

A final note of warning: We assumed that the only significant contribution to the integral came from the immediate vicinity of the saddle point $z = z_0$. This condition must be checked for each new problem.

Example 12.7.1 ASYMPTOTIC FORM OF THE GAMMA FUNCTION

In many physical problems, particularly in the field of statistical mechanics, it is desirable to have an accurate approximation of the gamma or factorial function of very large numbers. As listed in Table 1.2, the factorial function may be defined by the Euler integral

$$t! = \Gamma(t+1) = \int_0^\infty \rho^t e^{-\rho} d\rho = t^{t+1} \int_0^\infty e^{t(\ln z - z)} dz. \quad (12.109)$$

Here we have made the substitution $\rho = zt$ in order to convert the integral to the form given in Eq. (12.108). As before, we assume that t is real and positive, from which it follows that the integrand vanishes at the limits 0 and ∞ . By differentiating the exponent, which we call $w(z, t)$, we obtain

$$\frac{dw}{dz} = t \frac{d}{dz}(\ln z - z) = \frac{t}{z} - t, \quad w'' = -\frac{t}{z^2},$$

which shows that the point $z = 1$ is a saddle point and $\arg w''(1, t) = \arg(-t) = \pi$. Applying Eq. (12.106), we see that the direction of travel through the saddle point is

$$\theta = -\frac{\arg w''}{2} + \left(\frac{\pi}{2} \text{ or } \frac{3\pi}{2} \right) = 0 \text{ or } \pi;$$

the choice $\theta = 0$ is that consistent with deformation from a path that was originally along the real axis. In fact, what we have found is that the direction of steepest descent is along the real axis, a conclusion that we might have reached more or less intuitively.

Direct substitution into Eq. (12.108) with $g = t^{t+1}$, $F = e^{-t}$, $\theta = 0$, and $|w''| = -t$ yields

$$t! = \Gamma(t+1) \approx \sqrt{\frac{2\pi}{t}} t^{t+1} e^{-t} = \sqrt{2\pi} t^{t+1/2} e^{-t}. \quad (12.110)$$

This result is the leading term in Stirling's expansion of the gamma function. The method of steepest descents is probably the easiest way of obtaining this term. Further terms in the asymptotic expansion are developed in Section 13.4.

In this example the calculation was carried out assuming t to be real. This assumption is not necessary. We may show (Exercise 12.7.3) that Eq. (12.110) also holds when t is complex, provided only that its real part be required to be large and positive. ■

Sometimes the application of the saddle point method to a real integral results in a contour that goes through a saddle point that is not on the real axis. Here is a relatively simple example. A more complicated case of practical importance appears in the chapter on Bessel functions (see Section 14.6).

Example 12.7.2 SADDLE POINT METHOD AVOIDS OSCILLATIONS

As a second example of the method of steepest descents, consider the integral

$$H(t) = \int_{-\infty}^{\infty} \frac{e^{-t(z^2-1/4)} \cos tz}{1+z^2} dz, \quad (12.111)$$

which we wish to evaluate for large positive t . When t is large, the integrand oscillates very rapidly, and ordinary quadrature methods become difficult. We proceed by bringing $H(t)$ to a form appropriate for applying the saddle point method, replacing $\cos tz$ by $\cos tz + i \sin tz = e^{itz}$ (a replacement that does not change the value of the integral because we added an odd term to the previously even integrand). We then have

$$H(t) = \int_C g(z) e^{-t(z^2-iz-1/4)} dz, \quad (12.112)$$

with $g(z) = 1/(1+z^2)$. This form corresponds to $w(z) = -t(z^2 - iz - \frac{1}{4})$, so we have

$$w'(z) = -t(2z - i), \quad \text{which has a zero at } z_0 = i/2. \quad (12.113)$$

Then, at z_0 , which is a saddle point,

$$w_0 = 0, \quad w''(z_0) = -2t, \quad g(z_0) = \frac{4}{3}. \quad (12.114)$$

We also need the phase θ of the steepest-descent direction. Noting that $\arg(w''(z_0)) = \pi$ and applying Eq. (12.106), we find $\theta = 0$ (or π).

We are now ready to apply Eq. (12.108). The result is

$$H(t) \approx \frac{\sqrt{2\pi}(4/3)(e^0)}{|-2t|} = \frac{4}{3} \sqrt{\frac{\pi}{t}}. \quad (12.115)$$

As a check, we compare this approximate formula for $H(t)$ with the result of a tedious numerical integration: For $t = 100$, $H_{\text{exact}} = 0.23284$, and $H_{\text{saddle}} = 0.23633$. ■

Exercises

We present here a rather small number of exercises on the method of steepest descents. Several additional exercises appear elsewhere in this book, in particular in Section 14.6, where the technique is applied to the contour integral representations of Bessel functions.

- 12.7.1** Prove Jensen's theorem (that $|F(z)|^2$ can have no extremum in the interior of a region in which F is analytic) by showing that the mean value of $|F|^2$ on a circle about any

point z_0 is equal to $|F(z_0)|^2$. Explain why you can then conclude that there cannot be an extremum of $|F|$ at z_0 .

- 12.7.2** Find the steepest path and leading asymptotic expansion for the Fresnel integrals $\int_0^s \cos x^2 dx$, $\int_0^s \sin x^2 dx$.

Hint. Use $\int_0^1 e^{itz^2} dz$.

- 12.7.3** Show that the formula

$$\Gamma(1+s) \approx \sqrt{2\pi s} s^s e^{-s}$$

holds for complex values of s (with $\Re(s)$ large and positive).

Hint. This involves assigning a phase to s and then demanding that $\Im m[sf(z)]$ be constant in the vicinity of the saddle point.

12.8 DISPERSION RELATIONS

The concept of dispersion relations entered physics with the work of Kronig and Kramers in optics. The name **dispersion** comes from optical dispersion, a result of the dependence of the index of refraction on wavelength, or angular frequency. As we shall soon see, the index of refraction n may have a real part determined by the phase velocity and a (negative) imaginary part determined by the absorption. Kronig and Kramers showed in 1926–1927 that the real part of $(n^2 - 1)$ could be expressed as an integral of the imaginary part. Generalizing this, we shall apply the label **dispersion relations** to any pair of equations giving the real part of a function as an integral of its imaginary part and the imaginary part as an integral of its real part (we develop this in more detail below). The existence of such integral relations might be suspected as an integral analog of the Cauchy-Riemann differential equations, Eq. (11.9).

The applications in modern physics are widespread. For instance, the real part of the function might describe the forward scattering of a gamma ray in a nuclear Coulomb field (a dispersive process). Then the imaginary part would describe the electron-positron pair production in that same Coulomb field (the absorptive process). As will be seen later, the dispersion relations may be taken as a consequence of causality and therefore are independent of the details of the particular interaction.

We consider a complex function $f(z)$ that is analytic in the upper half-plane and on the real axis. We also require that $f(z)$ approach zero for large $|z|$ in the upper half-plane sufficiently rapidly that its integral over the semicircular part of the contour in Fig. 12.6 will be negligible. The point of these conditions is that we may express $f(z)$ by the Cauchy integral formula, Eq. (11.30), using this contour, obtaining

$$f(z_0) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{f(x)}{x - z_0} dx. \quad (12.116)$$

The integral over the contour shown in Fig. 12.6 has become an integral along the x -axis.

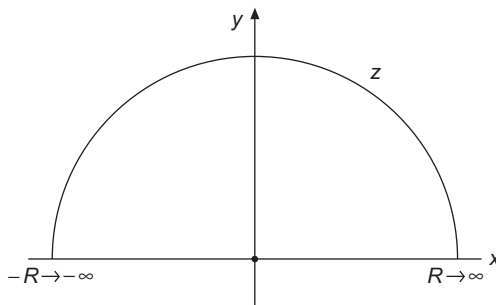


FIGURE 12.6 Contour for dispersion integral.

Equation (12.116) assumes that z_0 is in the upper half-plane, interior to the closed contour. If z_0 were in the lower half-plane, the integral would yield zero by the Cauchy integral theorem, Section 11.3. Now, if we move z_0 onto the real axis (then calling it x_0) and pass it via a small clockwise semicircle s in the upper half-plane, the contour integral (which would contain no singularities) would have nonzero contributions corresponding to a Cauchy principal value integral **minus** half the usual contribution from the pole at x_0 , or

$$\begin{aligned} 0 &= \oint \frac{f(x)}{x - x_0} dx + \int_s \frac{f(z)}{z - x_0} dz \\ &= \oint \frac{f(x)}{x - x_0} dx - \pi i f(x_0), \end{aligned}$$

equivalent to the final formula

$$f(x_0) = \frac{1}{\pi i} \oint_{-\infty}^{\infty} \frac{f(x)}{x - x_0} dx. \quad (12.117)$$

Note that the cut integral sign denotes the Cauchy principal value. Splitting Eq. (12.117) into real and imaginary parts⁶ yields

$$\begin{aligned} f(x_0) &= u(x_0) + i v(x_0) \\ &= \frac{1}{\pi} \oint_{-\infty}^{\infty} \frac{v(x)}{x - x_0} dx - \frac{i}{\pi} \oint_{-\infty}^{\infty} \frac{u(x)}{x - x_0} dx. \end{aligned}$$

Finally, equating real part to real part and imaginary part to imaginary part, we obtain

$$\begin{aligned} u(x_0) &= \frac{1}{\pi} \oint_{-\infty}^{\infty} \frac{v(x)}{x - x_0} dx, \\ v(x_0) &= -\frac{1}{\pi} \oint_{-\infty}^{\infty} \frac{u(x)}{x - x_0} dx. \end{aligned} \quad (12.118)$$

⁶The second argument, $y = 0$, is dropped: $u(x_0, 0) \rightarrow u(x_0)$.

These are the **dispersion relations**. The real part of our complex function is expressed as an integral over the imaginary part. The imaginary part is expressed as an integral over the real part. Alternatively, the real part can be called an integral transform of the imaginary part (and vice versa); the particular transform involved is known as a **Hilbert transform**, and we note that (apart from a minus sign) the Hilbert transform is its own inverse. Note that these relations are meaningful only when $f(x)$ is a complex function of the real variable x . Compare [Exercise 12.8.1](#).

From a physical point of view $u(x)$ and/or $v(x)$ may represent some physical measurements. Then $f(z) = u(z) + iv(z)$ is an analytic continuation over the upper half-plane, with the value on the real axis serving as a boundary condition.

Symmetry Relations

On occasion $f(x)$ will satisfy a symmetry relation and the integral from $-\infty$ to $+\infty$ may be replaced by an integral over positive values only. This is of considerable physical importance because the variable x might represent a frequency and only zero and positive frequencies are available for physical measurements. Suppose⁷

$$f(-x) = f^*(x). \quad (12.119)$$

Then

$$u(-x) + iv(-x) = u(x) - iv(x). \quad (12.120)$$

The real part of $f(x)$ is even and the imaginary part is odd.⁸ In quantum mechanical scattering problems these relations, [Eq. \(12.120\)](#), are called **crossing conditions**. To exploit these crossing conditions, we rewrite the first of [Eqs. \(12.118\)](#) as

$$u(x_0) = \frac{1}{\pi} \int_{-\infty}^0 \frac{v(x)}{x - x_0} dx + \frac{1}{\pi} \int_0^{\infty} \frac{v(x)}{x - x_0} dx. \quad (12.121)$$

Letting $x \rightarrow -x$ in the first integral on the right-hand side of [Eq. \(12.121\)](#) and substituting $v(-x) = -v(x)$ from [Eq. \(12.120\)](#), we obtain

$$\begin{aligned} u(x_0) &= \frac{1}{\pi} \int_0^{\infty} v(x) \left(\frac{1}{x + x_0} + \frac{1}{x - x_0} \right) dx \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{xv(x)}{x^2 - x_0^2} dx. \end{aligned} \quad (12.122)$$

⁷This is not just a curiosity. It ensures that the Fourier transform of $f(x)$ will be real. Or conversely, [Eq. \(12.119\)](#) is a consequence when $f(x)$ is obtained as the Fourier transform of a real function.

⁸ $u(x, 0) = u(-x, 0)$, $v(x, 0) = -v(-x, 0)$. Compare these symmetry conditions with those that follow from the Schwarz reflection principle, [Section 11.10](#).

Similarly,

$$v(x_0) = -\frac{2}{\pi} \int_0^{\infty} \frac{x_0 u(x)}{x^2 - x_0^2} dx. \quad (12.123)$$

The original Kronig-Kramers optical dispersion relations were in this form. The asymptotic behavior ($x_0 \rightarrow \infty$) of Eqs. (12.122) and (12.123) lead to quantum mechanical **sum rules**. See Exercise 12.8.4.

Optical Dispersion

The function $\exp[i(kx - \omega t)]$ can describe an electromagnetic wave moving along the x -axis in the positive direction with velocity $v = \omega/k$; ω is the angular frequency, k the wave number or propagation vector, and $n = ck/\omega$, the index of refraction. From Maxwell's equations with electric permittivity ε and magnetic permeability unity, and using Ohm's law with conductivity σ , the propagation vector k for a dielectric becomes⁹

$$k^2 = \varepsilon \frac{\omega^2}{c^2} \left(1 + i \frac{4\pi\sigma}{\omega\varepsilon} \right). \quad (12.124)$$

The presence of the conductivity (which means absorption) causes k^2 to have an imaginary part. The propagation vector k (and therefore the index of refraction n) have become complex.

For poor conductivity ($4\pi\sigma/\omega\varepsilon \ll 1$) a binomial expansion yields

$$k = \sqrt{\varepsilon} \frac{\omega}{c} + i \frac{2\pi\sigma}{c\sqrt{\varepsilon}}$$

and

$$e^{i(kx - \omega t)} = e^{i\omega(x\sqrt{\varepsilon}/c - t)} e^{-2\pi\sigma x/c\sqrt{\varepsilon}},$$

an attenuated wave.

Returning to the general expression for k^2 , Eq. (12.124), we find that the index of refraction becomes

$$n^2 = \frac{c^2 k^2}{\omega^2} = \varepsilon + i \frac{4\pi\sigma}{\omega}. \quad (12.125)$$

We take n^2 to be a function of the **complex** variable ω (with ε and σ depending on ω). However, n^2 does not vanish as $\omega \rightarrow \infty$ but instead approaches unity. It therefore does not satisfy the condition needed for a dispersion relation, but this difficulty can be circumvented by working with $f(\omega) = n^2(\omega) - 1$. The Kronig-Kramers relations then take

⁹See J. D. Jackson, *Classical Electrodynamics*, 3rd ed. New York: Wiley (1999), Sections 7.7 and 7.10. Equation (12.124) is in Gaussian units.

the form

$$\begin{aligned}\Re[n^2(\omega_0) - 1] &= \frac{2}{\pi} \int_0^\infty \frac{\omega \Im[n^2(\omega) - 1]}{\omega^2 - \omega_0^2} d\omega, \\ \Im[n^2(\omega_0) - 1] &= -\frac{2}{\pi} \int_0^\infty \frac{\omega_0 \Re[n^2(\omega) - 1]}{\omega^2 - \omega_0^2} d\omega.\end{aligned}\tag{12.126}$$

Knowledge of the absorption coefficient at all frequencies specifies the real part of the index of refraction, and vice versa.

The Parseval Relation

When the functions $u(x)$ and $v(x)$ are Hilbert transforms of each other, given by Eqs. (12.118), and each is square integrable,¹⁰ the two functions satisfy the scaling condition

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} |v(x)|^2 dx.\tag{12.127}$$

This is the Parseval relation.

To derive Eq. (12.127), we start with

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} dx \left[\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{v(s) ds}{s - x} \right] \left[\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{v(t) dt}{t - x} \right],$$

using the formula for $u(x)$ from Eq. (12.118) twice. Integrating first with respect to x , we have

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} v(s) ds \int_{-\infty}^{\infty} v(t) dt \frac{1}{\pi^2} \int_{-\infty}^{\infty} \frac{dx}{(s - x)(t - x)},\tag{12.128}$$

where both principal-value limits at the singularities of the integrand must now be taken for the x integration. As shown in Exercise 12.8.8, that integration yields a delta function¹¹:

$$\frac{1}{\pi^2} \int_{-\infty}^{\infty} \frac{dx}{(s - x)(t - x)} = \delta(s - t).$$

Thus,

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} v(t) dt \int_{-\infty}^{\infty} v(s) \delta(s - t) ds.\tag{12.129}$$

¹⁰This means that $\int_{-\infty}^{\infty} |u(x)|^2 dx$ and $\int_{-\infty}^{\infty} |v(x)|^2 dx$ are finite.

¹¹Note that when $s = t$, the integrand has the same sign (for small ε) at $x = s - \varepsilon$ and at $x = s + \varepsilon$, so the limit defining the principal value then does not exist. The singularity in the integration is that which is needed to represent a delta function.

Then the s integration is carried out by inspection, using the defining property of the delta function:

$$\int_{-\infty}^{\infty} v(s)\delta(s-t)ds = v(t). \quad (12.130)$$

Substituting Eq. (12.130) into Eq. (12.129), we have Eq. (12.127), the Parseval relation. Again, in terms of optics, the presence of refraction over some frequency range ($n \neq 1$) implies the existence of absorption, and vice versa.

Exercises

- 12.8.1** Assume that the function $f(z)$ satisfies the conditions for the dispersion relations. In addition, assume that $f(z) = f^*(z^*)$, i.e., that it meets the conditions of the Schwarz reflection principle, Eq. (11.127). Show that $f(z)$ is identically zero.
- 12.8.2** For $f(z)$ such that we may replace the closed contour of the Cauchy integral formula by an integral over the real axis we have

$$f(x_0) = \frac{1}{2\pi i} \left\{ \int_{-\infty}^{x_0-\delta} \frac{f(x)}{x-x_0} dx + \int_{x_0+\delta}^{\infty} \frac{f(x)}{x-x_0} dx \right\} + \frac{1}{2\pi i} \int_C \frac{f(x)}{x-x_0} dx.$$

Here we take C to be a small semicircle about x_0 in the lower half-plane. Show that the formula for $f(x_0)$ reduces to

$$f(x_0) = \frac{1}{\pi i} \int_{-\infty}^{\infty} \frac{f(x)}{x-x_0} dx,$$

which is Eq. (12.117).

- 12.8.3** (a) The function $f(z) = e^{iz}$ does not vanish at the endpoints of the range of $\arg z$, a and π . Show, with the help of Jordan's lemma, Eq. (11.102), that Eq. (12.116) still holds.
- (b) For $f(z) = e^{iz}$ verify by direct integration the dispersion relations, Eq. (12.117) or Eqs. (12.118).
- 12.8.4** With $f(x) = u(x) + iv(x)$ and $f(x) = f^*(-x)$, show that as $x_0 \rightarrow \infty$,

- (a) $u(x_0) \sim -\frac{2}{\pi x_0^2} \int_0^{\infty} x v(x) dx,$
- (b) $v(x_0) \sim \frac{2}{\pi x_0} \int_0^{\infty} u(x) dx.$

In quantum mechanics relations of this form are often called **sum rules**.

- 12.8.5 (a) Given the integral equation (valid for all real x_0)

$$\frac{1}{1+x_0^2} = \frac{1}{\pi} \oint_{-\infty}^{\infty} \frac{u(x)}{x-x_0} dx,$$

use Hilbert transforms to determine $u(x_0)$.

- (b) Verify that the $u(x_0)$ found as your answer to part (a) actually satisfies the integral equation.
 (c) From $f(z)|_{y=0} = u(x) + iv(x)$, replace x by z and determine $f(z)$. Verify that the conditions for the Hilbert transforms are satisfied.
 (d) Are the crossing conditions satisfied?

$$\text{ANS. (a) } u(x_0) = \frac{x_0}{1+x_0^2}, \quad \text{(c) } f(z) = (z+i)^{-1}.$$

- 12.8.6 (a) If the real part of the complex index of refraction (squared) is constant (no optical dispersion), show that the imaginary part is zero (no absorption).
 (b) Conversely, if there is absorption, show that there must be dispersion. In other words, if the imaginary part of $n^2 - 1$ is not zero, show that the real part of $n^2 - 1$ is not constant.

- 12.8.7 Given $u(x) = x/(x^2 + 1)$ and $v(x) = -1/(x^2 + 1)$, show by direct evaluation of each integral that

$$\int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} |v(x)|^2 dx.$$

$$\text{ANS. } \int_{-\infty}^{\infty} |u(x)|^2 dx = \int_{-\infty}^{\infty} |v(x)|^2 dx = \frac{\pi}{2}.$$

- 12.8.8 Take $u(x) = \delta(x)$, a delta function, and **assume** that the Hilbert transform equations hold.

- (a) Show that

$$\delta(w) = \frac{1}{\pi^2} \oint_{-\infty}^{\infty} \frac{dy}{y(y-w)}.$$

- (b) With changes of variables $w = s - t$ and $x = s - y$, transform the δ representation of part (a) into

$$\delta(s-t) = \frac{1}{\pi^2} \oint_{-\infty}^{\infty} \frac{dx}{(x-s)(x-t)}.$$

Note. The δ function is discussed in Section 1.11.

Additional Readings

Abramowitz, M., and I. A. Stegun, eds., *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables* (AMS-55). Washington, DC: National Bureau of Standards (1972), reprinted, Dover (1974).

Lewin, L., *Polylogarithms and Associated Functions*. New York: North-Holland (1981). This is a definitive resource for the dilogarithm and its generalizations up through its publication date. It is clear and no more difficult than necessary.

McBride, E. B., *Obtaining Generating Functions*. New York: Springer-Verlag (1971). An introduction to methods of obtaining generating functions, both for sets of functions arising from ODEs and for those that do not.

Nussenzveig, H. M., *Causality and Dispersion Relations*, Mathematics in Science and Engineering Series, Vol. 95. New York: Academic Press (1972). This is an advanced text covering causality and dispersion relations in the first chapter and then moving on to develop implications in a variety of areas of theoretical physics.

Talman, J. D., *Special Functions*. New York: W. A. Benjamin (1968). Develops the theory of a number of special functions using their underlying group-theoretical properties, including presentation of their generating functions.

Wyld, H. W., *Mathematical Methods for Physics*. Reading, MA: Benjamin/Cummings (1976), Perseus Books (1999). This is a relatively advanced text that contains an extensive discussion of dispersion relations.